CHAPTER 1

INTRODUCTION

In the digital age, memes have evolved into a powerful form of communication, blending humor, sarcasm, and cultural commentary into compact visual-textual formats. Unlike traditional reviews or surveys, memes often carry nuanced emotional signals that reflect public sentiment toward products, services, and brands. However, these signals are difficult to decode using conventional sentiment analysis tools, which typically focus on textual data alone. This limitation creates a gap in understanding consumer emotions expressed through multimodal content

The "Memesent Review Engine" addresses this challenge by introducing a novel framework for multimodal sentiment analysis. By leveraging advanced machine learning techniques—such as transformer-based models for text and convolutional neural networks (CNNs) for images—the system captures subtle emotional cues embedded in meme content. This dual-modality approach enables a more holistic understanding of consumer feedback, especially in contexts where humor,irony, or sarcasm play a central role.

The core objective of this project is to transform unstructured, user- generated meme content into actionable business insights. By analyzing memes shared across platforms like Instagram, Twitter, and Facebook, the system can detect early signs of customer satisfaction or dissatisfaction, track sentiment trends, and evaluate the emotional impact of marketing campaigns. This real-time feedback loop empowers businesses to make data-driven decisions in product development, customer engagement, and brand positioning.

In addition to analyzing meme-based sentiments, the proposed MemeSent Review Engine contributes to the advancement of multimodal artificial intelligence by bridging the gap between human-like perception and computational interpretation. Unlike conventional text-only models, this system accounts for the synergy between visual and linguistic elements, allowing a deeper and more authentic understanding of online discourse. By combining textual semantics and visual context, it deciphers complex emotional signals that often influence consumer perception subconsciously.

Moreover, the system plays an essential role in modern marketing and communication strategies, where memes serve as cultural indicators of public opinion. Businesses, political organizations, and social platforms can use the system to monitor audience engagement, detect emerging trends, and predict public response to campaigns. The integration of Natural Language Processing (NLP) and Computer Vision (CV) enhances its predictive capabilities, making it suitable for analyzing large-scale datasets in real time.

This framework not only assists in identifying audience sentiment but also aids in strategic decision-making by providing visual analytics and data-driven insights. The ability to process multimodal data gives the system a competitive edge in domains such as brand monitoring, entertainment analysis, digital journalism, and psychological research. Overall, the MemeSent Review Engine represents a forward-looking step toward intelligent, context-aware sentiment analysis that mirrors human emotional understanding in the digital landscape.

1.1 IDENTICATION OF THE PROBLEMS:

1.1.1 PROBLEM STATEMENT

Limitations of Traditional Sentiment Analysis

Traditional sentiment analysis tools primarily rely on textual data and struggle to interpret the emotional depth embedded in memes, which often combine humor, sarcasm, and visual cues. This leads to inaccurate sentiment classification and missed insights, especially in consumer feedback shared through social media. There is a need for a multimodal system that can decode both text and image components to capture true consumer emotions.

Unstructured Feedback in Meme Culture

Modern consumers increasingly express product opinions through memes rather than structured reviews or surveys. These memes are rich in emotional and cultural context but are difficult to analyze due to their informal and multimodal nature. Businesses lack tools to convert this unstructured meme content into actionable feedback, limiting their ability to respond to customer sentiment in real time.

Missed Marketing Opportunities

Brands often fail to detect early signs of public dissatisfaction or praise because existing sentiment tools overlook the emotional signals in memes. Without a system that can accurately classify meme-based sentiment, companies risk missing viral trends, misjudging campaign impact, and losing opportunities to refine messaging and boost engagement.

Need for Emotionally Intelligent AI

Current AI models used for sentiment analysis are not equipped to handle the fusion of textual and visual data found in memes. They lack the emotional intelligence to detect irony, satire, and subtle visual metaphors. This limits their effectiveness in understanding consumer behavior and guiding emotionally resonant product and marketing decisions.

1.2 OBJECTIVE OF THE PROJECT:

1.2.1 To Design a Multimodal Sentiment Analysis Framework:

The primary objective of this project is to develop a robust AI-based system that can analyze both textual and visual elements of memes to classify sentiment accurately. Unlike traditional sentiment analysis tools that rely solely on text, this system integrates image processing and natural language understanding to capture the full emotional context of meme content. By doing so, it addresses the limitations of unimodal models and enables a more comprehensive interpretation of consumer feedback.

1.2.2 To Extract and Fuse Emotional Signals from Text and Images:

Memes often convey complex emotions through a combination of sarcastic captions and expressive visuals. This project aims to extract meaningful features from both modalities—using techniques like TF-IDF and BERT for text, and CNNs and color histograms for images—and fuse them into a unified representation. This fusion allows the model to detect subtle emotional cues such as irony, humor, or dissatisfaction that are otherwise lost in single-modality analysis.

1.2.3 To Enable Real-Time Consumer Sentiment Monitoring for Business Intelligence:

Another key objective is to empower businesses with real-time insights into public perception of their products, services, and campaigns. By analyzing meme-based feedback from social media platforms, the system can identify emerging sentiment trends, detect viral praise or backlash, and provide early warnings for brand reputation management. This allows marketing and product teams to respond proactively and make data-driven decisions that align with consumer emotions.

1.2.4 To Drive Product Innovation and Sales Growth through Emotion-Aware Analytics:

Ultimately, the Memesent Review Engine is designed to bridge the gap between consumer expression and business strategy. By decoding the emotional undercurrents in meme content, the system helps companies refine product features, personalize marketing messages, and enhance customer engagement. The long-term goal is to integrate this sentiment intelligence into broader business analytics pipelines, supporting strategic planning, demand forecasting, and emotionally resonant brand storytelling

1.3 SIGNIFICANCE:

1.3.1 Bridge the Gap Between Consumer Expression and Business Insight

In the age of social media, memes have become a dominant form of consumer expression—often more authentic and emotionally charged than traditional reviews or surveys. This project recognizes the untapped potential of meme content and introduces a system that decodes both textual and visual sentiment. By doing so, it enables businesses to understand how customers truly feel about their products, services, and campaigns in a culturally relevant format.

1.3.2 Advancing Sentiment Analysis Through Multimodal AI

The Memesent Review Engine goes beyond conventional sentiment analysis by integrating machine learning models that process both text and image data. This multimodal approach allows for the detection of complex emotional signals such as sarcasm, irony, and humor—elements that are often missed by text-only models. The project contributes to the evolution of AI in understanding human emotion, making sentiment analysis more accurate and context-aware.

1.3.3 Empowering Emotion-Driven Decision Making

By classifying meme-based feedback into positive, negative, or neutral categories, the system provides actionable insights that can guide product development, marketing strategies, and customer engagement efforts. Businesses can use this data to refine messaging, respond to public sentiment in real time, and create emotionally resonant campaigns that boost brand loyalty and sales performance.

1.3.4 Enhancing Real-Time Market Responsiveness

The ability to monitor meme sentiment across platforms like Instagram and Twitter offers a dynamic feedback loop for brands. This real-time responsiveness helps companies detect viral trends, identify emerging issues, and adapt quickly to consumer needs. The project thus plays a critical role in strengthening brand reputation and maintaining competitive advantage in fast-moving digital markets.

1.4 EXISTING SYSTEM:

1.4.1 Text-Only Sentiment Analysis Models

Most existing sentiment analysis systems are designed to process structured textual data such as reviews, tweets, or comments. These models include:

a) NaiveBayes: A probabilistic classifier that assumes feature independence. While it's fast and simple, it fails to capture contextual meaning, sarcasm, or complex sentence structures—especially in meme captions that often rely on cultural nuance.

b) Support Vector Machine (SVM): A robust classifier that performs well with engineered features. However, it depends heavily on manual preprocessing and lacks adaptability to informal, slang-heavy, or humorous text.

c)Long Short-Term Memory (LSTM): A deep learning model capable of understanding sequential data and context. Though more powerful than traditional models, LSTM is computationally expensive and still limited to textual inputs unless extended with multimodal capabilities**.**

1.4.2 Image-Only Sentiment Analysis Models

Some Systems attempt to analyze sentiment from images alone ,using techniques like :

a) Convolutional Neural Networks (CNNs): These models extract visual features such facial expressions, color tones, and object presence. While useful for detecting emotions like happiness or anger in faces, they fall short when the image is abstract, symbolic, or paired with emotionally charged text.

b) Color Histograms and Texture Analysis: These low-level features can hint at mood (e.g., dark tones for sadness), but they lack semantic understanding and cannot interpret visual metaphors or cultural references.

1.4.3 Unimodel Limitations in Real-World Applications

a)Lack of Fusion: Existing systems rarely combine text and image features. This unimodal This unimodal approach leads to incomplete sentiment detection,especially in memes where meaning arises from the interplay caption

b)No Real-Time Adaptability: Traditional systems are not designed to monitor fast-changing meme trends on platforms like Instagram or Twitter. They lack scalability and responsiveness to viral content**.**

c) Limited Business Utility: These systems do not translate meme sentiment into actionable business insights. They fail to link emotional feedback to product perception, campaign performance, or sales impact.

1.5 PROPOSED SYSTEM:

1.5.1 Multimodel Sentiment Anlysis Integration

The proposed system introduces a multimodal sentiment analysis framework that processes both textual and visual components of memes. Unlike traditional models that rely on a single modality, this system fuses features from captions and images to capture nuanced emotional signals such as sarcasm, irony, and humor. It leverages TF-IDF and BERT for text representation, and CNN-based color histograms for image features, enabling a richer understanding of consumer sentiment.

1.5.2 Feature Fusion and Random Forest Classification

To overcome the limitations of unimodal models, the proposed system performs horizontal stacking of text and image features to create a unified multimodal vector. This fused representation is used to train a Random Forest classifier, chosen for its robustness and ability to handle high-dimensional, mixed-type data. The model is optimized through hyperparameter tuning and evaluated using precision, recall, F1-score, and confusion matrix analysis.

1.5.3 Real-Time Sentiment prediction and Trend Monitoring

The system supports real-time sentiment prediction on newly collected memes from platforms like Instagram and Twitter. It processes incoming meme content through the same preprocessing pipeline and classifies sentiment as positive, negative, or neutral. These predictions are aggregated over time to identify emerging trends, viral reactions, and shifts in public perception—providing businesses with timely insights for campaign optimization and brand management.

1.5.4 Emotion-Aware Business Intelligence

Beyond classification, the proposed system translates meme sentiment into actionable business insights. It enables companies to decode emotional feedback, refine product features, and personalize marketing messages

By integrating sentiment outputs into analytics dashboards, the system supports strategic planning, demand forecasting, and emotionally resonant storytelling—bridging the gap between consumer expression and business strategy.

1.5.5 Scalability and Cultural Adaptability

The architecture is designed for scalability across diverse datasets and social platforms. It can be extended to support multilingual meme content using transformer-based models like ROBERTa and LASER embeddings. This adaptability ensures relevance across global markets and enhances the system's ability to interpret culturally embedded emotional signals.

1.6 INNOVATION IN SYSTEM DETECTION:

The Memesent Review Engine introduces a groundbreaking shift in sentiment analysis by moving beyond traditional text-only models and embracing a multimodal approach. This innovation lies in its ability to decode emotional signals from both textual captions and visual elements of memes something conventional systems fail to capture. By fusing these two modalities, the system can detect subtle tones like sarcasm, irony, humor, and cultural references, which are often embedded in meme formats and overlooked by standard classifiers.

Unlike existing models that rely on handcrafted features or shallow learning algorithms, this project leverages transformer-based architectures (e.g., BERT) for deep contextual understanding of text, and Convolutional Neural Networks (CNNs) for extracting emotional cues from images. These features are then combined using fusion techniques to create a unified sentiment representation, enabling more accurate classification into positive, negative, or neutral categories.

Another key innovation is the system's ability to process real-time meme data from social platforms, allowing businesses to monitor public sentiment as it evolves. This dynamic feedback loop supports emotion-aware decision-making, helping brands respond proactively to consumer reactions and optimize product messaging and campaign strategies. By integrating machine learning with cultural intelligence, the Memesent Review Engine transforms informal, user-generated content into structured insights. This not only enhances sentiment detection accuracy but also opens new avenues for emotion-driven analytics, making it a pioneering tool in the field of AI-powered consumer intelligence.

1.7 APPLICATIONS OF THE PROJECT

The Meme Sent Review Engine offers broad applicability across multiple industries, research domains, and digital ecosystems. Its unique capability to analyze both textual and visual data allows it to generate accurate, real-time insights about public emotions, making it a valuable asset in the era of social media-driven marketing.

One of the key applications of this project lies in digital marketing and brand management. Organizations can monitor public reactions to advertisements, campaigns, or newly launched products by analyzing memes circulating on social media platforms. The system identifies the prevailing emotional tone—positive, negative, or neutral—helping companies evaluate the effectiveness of marketing strategies and tailor future campaigns accordingly.

In the e-commerce sector, the project can be used for product feedback analysis, allowing businesses to identify customer satisfaction trends, address concerns, and predict consumer preferences based on meme-driven engagement. Similarly, in sales analytics, the correlation between meme sentiment and product sales data provides a predictive mechanism to estimate future market performance.

The system also has potential in social media monitoring and public opinion research, where it can be employed to study large-scale sentiment patterns during social events, political campaigns, or cultural trends. Furthermore, it supports academic and research applications in the fields of Natural Language Processing (NLP), Computer Vision (CV), and Artificial Intelligence (AI) by serving as a reference model for multimodal sentiment detection and feature fusion methodologies.

From an industrial perspective, the engine can be integrated with real-time data visualization dashboards, CRM systems, and social listening platforms to offer continuous insight into user engagement and emotional response. Its scalable design makes it suitable for cloud-based deployment, enabling high-volume data processing and cross-platform analytics.

Overall, the Meme Sent Review Engine bridges the gap between human emotion and computational understanding, transforming unstructured meme content into actionable intelligence. By providing valuable insights into consumer psychology and digital behavior, the system supports decision-making in marketing, entertainment, research, and data analytics industries, making it a robust and forward-looking technological innovation.

1.8 PROJECT OUTCOMES

The Meme Sent Review Engine project has achieved its primary objectives by successfully designing and implementing a system capable of analyzing and classifying the emotional sentiment of memes using machine learning and deep learning techniques. The project effectively integrates image processing and natural language processing (NLP) to understand both the visual and textual components of memes, thereby enhancing the accuracy of sentiment detection.

Through systematic experimentation and model training, the proposed system demonstrated the ability to classify memes into positive, negative, or neutral categories with high precision. The performance evaluation confirmed that the model can efficiently interpret emotions, sarcasm, and humor from multimodal data, overcoming the limitations of traditional text-only sentiment analysis.

The project outcomes highlight the practical applicability of AI-driven emotion recognition systems in various real-world domains such as social media analytics, digital marketing, brand monitoring, and political sentiment studies. By analyzing online meme content, organizations and researchers can now gain valuable insights into public perception, trends, and opinions in an automated manner.

- Additionally, the project provided valuable learning experiences in:
- Dataset collection and preprocessing of multimodal data (text + image)
- Model training using CNN and deep learning frameworks
- Integration of machine learning models with a Flask web interface
- Visualization of sentiment results for better understanding and usability

Overall, the Meme Sent Review Engine successfully demonstrates how artificial intelligence can interpret human emotions through digital memes, paving the way for advanced research in multimodal sentiment analysis and emotion-aware computing. The project not only met its technical goals but also contributed significantly to skill development in areas such as data science, programming, and AI system design. Furthermore, it enhanced understanding of how visual and textual elements can be fused for emotional context detection, promoting innovative approaches in human–computer interaction. This project serves as a foundation for future developments in emotion-driven AI systems, capable of transforming digital communication, social media monitoring, and intelligent marketing analytics.

1.9 WORKING PRINCIPLE

The working principle of the proposed system is centered on the concept of multimodal sentiment analysis, where both visual and textual features of memes are analyzed together to identify the emotional tone being conveyed. The system begins by collecting meme data from various social media sources such as Facebook, Instagram, Reddit, and Twitter, ensuring a diverse and representative dataset. Each meme in the dataset typically contains a combination of image and text, which together carry humor, sarcasm, or emotion. The first stage of the system involves preprocessing, where the text embedded in the meme image is extracted using Optical Character Recognition (OCR) technology. This OCR-generated text is then refined by performing several cleaning operations, including removal of unwanted symbols, correction of spelling errors, and normalization of capitalization and punctuation. The cleaned text is then processed through Natural Language Processing (NLP) techniques such as tokenization, stop-word removal, and lemmatization to make it suitable for feature extraction and further analysis. Parallel to the text analysis, the system performs visual analysis on the meme images. This is achieved using Convolutional Neural Networks (CNN), which are highly effective for image recognition tasks. The CNN extracts important visual features such as color tones, facial expressions, object presence, and background context, all of which contribute to understanding the emotion conveyed visually. The extracted visual and textual features are then transformed into numerical representations, allowing them to be combined into a single feature space. This fusion of multimodal features forms the foundation of the sentiment classification process, ensuring that both linguistic and visual cues are considered simultaneously.

Once the multimodal feature vector is obtained, it is passed into a sentiment classification model that predicts the emotional category of the meme. The model is trained on labeled data where each meme is associated with a sentiment label such as positive, neutral, or negative. Machine learning algorithms like Support Vector Machines (SVM) or deep learning-based classifiers such as CNN-LSTM or Transformer architectures can be used to perform this classification. These models learn to identify subtle relationships between text and image elements that together define the sentiment of the meme. For instance, a meme with a smiling face and humorous caption is classified as positive, whereas one with sarcastic text and dull color tones may fall under the negative category.The final stage involves displaying or storing the predicted sentiment, which can be used for various analytical purposes such as studying public opinion trends, understanding audience reactions, or filtering inappropriate content on social platforms. The system's design ensures that each module — from OCR text extraction to multimodal feature fusion — works in coordination to produce accurate sentiment predictions. The strength of this working principle lies in its ability to capture both the visual and textual dimensions of emotion, which are often interdependent in memes. By combining the cognitive understanding of language with the perceptual understanding of visuals, the system achieves higher accuracy and contextual awareness than traditional sentiment analysis methods that rely solely on text

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION:

In recent years, sentiment analysis has evolved from traditional text-based classification to sophisticated multimodal approaches that incorporate both textual and visual data. This shift is particularly relevant in the context of social media, where users increasingly express opinions through memes rich, culturally embedded artifacts that blend humor, sarcasm, and emotion. Conventional sentiment analysis techniques often fall short in decoding the nuanced emotional signals embedded in such multimodal content.

To address these limitations, researchers have explored advanced machine learning and deep learning models, including transformer-based architectures like BERT and RoBERTa for text, and convolutional neural networks (CNNs) for image analysis. The fusion of these modalities has led to more accurate and context-aware sentiment classification systems. Additionally, cross-lingual transfer learning and language-agnostic embeddings have enabled sentiment analysis across diverse linguistic landscapes, expanding the applicability of these models to global survey.

This literature survey reviews key contributions in the field of multimodal sentiment analysis, with a focus on meme-based emotion detection. It highlights the evolution of sentiment analysis, the integration of text and image features, the role of deep learning architectures, and the challenges unique to meme interpretation.These insights form the foundation for developing the proposed Memesent Review Engine, which aims to extract actionable consumer feedback from meme content to inform product strategy and drive sales growth. In recent years, sentiment analysis has evolved from traditional text-based classification to sophisticated multimodal approaches that incorporate both textual and visual data. This shift is particularly relevant in the context of social media, where users increasingly express opinions through memes—rich, culturally embedded artifacts that blend humor, sarcasm, and emotion. Conventional sentiment analysis techniques often fall short in decoding the nuanced emotional signals embedded in such multimodal content. To address these limitations, researchers have explored advanced machine learning and deep learning models, including transformer-based architectures like BERT and RoBERTa for text, and convolutional neural networks (CNNs) for image analysis. The fusion of these modalities has led to more accurate and context-aware sentiment classification systems. Additionally, cross-lingual transfer learning and language-agnostic embeddings have enabled sentiment analysis across diverse linguistic landscapes, expanding the applicability of these models to global.

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insights form the foundation for developing the proposed Memesent Review Engine, which aims to extract actionable consumer feedback from meme content to inform roduct strategy and drive sales grow.

A. Title : Cross-lingual Deep Neural Transfer Learning in Sentiment Analysis

Published by: Procedia Computer Science, Elsevier

Year: 2020

Introduction:

This paper introduces a novel approach to sentiment analysis using language-agnostic sentence embeddings. It addresses the challenge of analyzing sentiment in low-resource languages like Polish by leveraging transfer learning and deep neural networks.

System Overview:

The system uses LASER embeddings to create universal sentence representations. These are fed into CNN and BiLSTM models trained on the PolEmo 1.0 corpus. The trained model is then evaluated on MultiEmo-Test 1.0, a multilingual dataset, demonstrating effective cross-lingual sentiment classification.

Limitations:

The model struggles to accurately detect neutral (0) and weak-negative (WN) sentiments, with F1-scores close to 0.00. This shows that it has difficulty recognizing subtle emotional tones. One major reason for this could be the limited number of examples available for these sentiment classes, which affects its learning ability.

In cross-lingual testing, the model relies on translations of the PolEmo dataset (used in MultiEmo-Test 1.0). However, the quality of these translations might have influenced the results, and the paper does not deeply analyze this issue. Importantly, the transfer was tested only from Polish to other languages — the reverse direction or multi-source training setups were not explored. The system used LASER sentence embeddings for multilingual representation.

Other advanced models like mbret or XLM-R were not compared, which might have limited performance. Similarly, only CNN and BiLSTM classifiers were used, without evaluating transformer-based architectures that typically perform better in sentiment analysis. Performance also varied across different domains such as Hotels, Medicine, School, and Products, suggesting that the model might not generalize well to unseen domains.

Although designed for low-resource languages, the model was trained using only Polish, so it may not effectively generalize to other low-resource languages.Even the best-performing versions of the model achieved F1-scores below 0.4 (around 35.56%), indicating that it’s not yet ready for high-accuracy real-world applications.

Dataset Used	Methods / Algorithms	Implementation Summary	Outcome	Future Scope
PolEmo 1.0 (Polish) and MultiEmo-Test 1.0 (translated into 8 languages)	LASER embeddings, CNN, BiLSTM	Language- agnostic sentence embeddings used to train models on Polish data and test on multilingual corpora	BiLSTM with LASER outperformed CNN in 65% of cases; effective cross-lingual sentiment transfer	Extend to more languages and domains; integrate with real-time multilingual feedback systems

Table : 1 Comparative Overview of Sentiment Analysis Methods Using Multilingual Datasets

B. Title: Sentiment Analysis of Movie Reviews: A Flask Application Using CNN with Robert Embeddings

Researchers: Biplov Paner , Bipul Thapa, Bish wash Paner

Published by: Systems and Soft Computing, Elsevier

Year: 2025

Introduction:

This study proposes a hybrid deep learning model for sentiment analysis of IMDB movie reviews. It combines Robert's contextual embeddings with CNN to improve sentiment classification accuracy.

System Overview:

The model preprocesses text data, generates embeddings using Robert, and extracts features via CNN. It achieves 91.5% test accuracy and is deployed as a Flask application for real-time sentiment prediction.

Limitations:

Limited Model Generalization : Although the R-CNN model performs well on the IMDB dataset, it appears to be overfitted to that specific dataset's writing style and vocabulary. Its ability to handle data from other domains or informal sources such as social media posts remains uncertain.

No Sentiment Intensity or Emotion Detection: The system classifies reviews only as positive or negative, without identifying the strength of emotions, sarcasm, or subtle contextual cues. As a result, it cannot capture deeper emotional variations present in real-world text.

3.Limited Flask Application Scope: The Flask-based web interface provides a simple demonstration of sentiment prediction but lacks advanced features such as visualizing sentiment trends, processing multiple reviews at once, or handlingreal-time data streams.

Lack of Explainability:While the model's accuracy is highlighted, the study does not discuss why the model makes certain predictions. This absence of interpretability reduces the system's transparency and user trust.

NarrowModelComparison**:**Performance evaluation is limited to comparisons with Logistic Regression and LSTM baselines. Modern transformer-based models like BERT, DistilBERT, or XLNet were not included, leaving uncertainty about how the R-CNN would perform against state-of-the-art alternatives.

FutureWorkDependencies:The paper acknowledges that future work will focus on adding transformer-based attention mechanisms and benchmarking against commercial APIs such as Google Cloud and IBM Watson. This suggests that the current version serves primarily as a foundational prototype.

Limited DatasetDomain:Training and testing were conducted only on the IMDB movie reviews dataset (around 50,000 samples). Consequently, the model's effectiveness on other forms of text—such as tweets, blogs, or multilingual reviews has not been evaluated.Binary Sentiment Classification Only:The model handles only two sentiment categories: positive and negative. Real-world data often include neutral or mixed opinions, which are not addressed in this approach.

LanguageRestriction:Since the system is developed exclusively for English text, it cannot process multilingual or code-mixed content. This limits its usability for global audiences. HighComputationalCost:The use of RoBERTa embeddings significantly increases GPU and memory requirements. This makes real-time prediction and large-scale deployment challenging, especially in resource-limited environments.

Dataset Used	**Methods / Algorithms**	**Implementation Summary**	**Outcome**	**Future Scope**
IMDB Movie Review Dataset (~50,000 entries)	RoBERTa embeddings, CNN, Flask deployment	Preprocessed text, generated contextual embeddings, trained CNN model, deployed via Flask for real-time prediction	Achieved 91.5% test accuracy; robust performance across varied review lengths	Expand to multimodal inputs (e.g., trailers, posters); integrate with recommendation engines

Table:2 Real-Time Sentiment Analysis Using Robert a Embeddings on IMDB Movie Review

C.Title: Sentiment Analysis on Twitter Data
Researchers: Varsha Sahayak, Vijaya Shete, Apashabi Pathan
Published by: International Journal of Innovative Research in Advanced Engineering (IJIRAE)
Year: 2024

Introduction:

This paper explores sentiment classification of Twitter data using traditional machine learning models. It emphasizes the importance of preprocessing and feature engineering for short-text sentiment analysis.

System Overview:

The system applies Naive Bayes, MaxEnt, and SVM classifiers on preprocessed Twitter data. Techniques like unigram modeling and tree kernels are used to enhance classification accuracy across positive, negative, and neutral sentiments.

Limitations:

Short Text Length of Tweets:Tweets are limited to 140–148 characters, making it difficult to extract complete contextual meaning and emotional depth compared to full-length reviews.

Ambiguity and Mixed Sentiments:Tweets often contain multiple emotions or opinions within a single sentence (e.g., "The meal was awesome but the service was terrible"), which causes confusion for classifiers.

Domain Dependency:Sentiment analysis models trained on Twitter data cannot be directly applied to other platforms like Facebook or blogs because of differences in text structure and vocabulary.

Noisy and Informal Language:Twitter data includes acronyms, emoticons, hashtags, abbreviations, and spelling errors, making preprocessing and classification challenging.

Limited Feature Set:The study primarily focuses on unigram, feature-based, and tree kernel models. These may not capture deeper semantic relations or contextual dependencies.

Dependence on Lexical Resources:The proposed approach relies heavily on sentiment dictionaries and prior polarity scores, which may not effectively handle new or evolving slang and internet language.

Negation and Sarcasm Handling:Although negation is considered, sarcasm and irony common in tweets are not effectively addressed, leading to possible misclassification.

Language and Cultural Limitations:The model is mainly applicable to English tweets; multilingual or code-mixed data are not analyzed.

Data Quality and Noise:Since distant supervision and emoticons are used as labels,

training data may contain mislabeled or irrelevant tweets, affecting model accuracy.

Limited Evaluation:The paper does not present quantitative performance metrics (e.g., accuracy, F1-score), so it is hard to assess real-world applicability.

Dataset Used	**Methods / Algorithms**	**Implementation Summary**	**Outcome**	**Future Scope**
Twitter dataset (tweets labeled as positive, negative, neutral)	Naive Bayes, MaxEnt, SVM, Unigram & Tree Kernel models	Applied traditional ML classifiers with preprocessing and feature engineering	Accurate classification of short-text sentiments; SVM showed strong performance	pply to multilingual tweets; enhance with transformer-based models for sarcasm detection

Table 3: Performance Comparison of Traditional Classification Algorithms on Twitter Sentiment Dataset

D.Title: The Social Media Sentiment Analysis Framework: Deep Learning for Sentiment Analysis on Social Media
Researchers: Prasanna Kumar Rangarajan, Bharathi Mohan Gurusamy, Gayathri Muthurasu, et al.
Published by: International Journal of Electrical and Computer Engineering (IJECE)
Year: 2024

Introduction:

This research proposes a comprehensive framework for sentiment analysis on social media platforms. It introduces a custom-built emotion dictionary and evaluates multiple deep learning models for sentiment extraction.

System Overview:

The SMSA framework includes data acquisition, preprocessing (tokenization, POS tagging, NER), and sentiment classification using deep neural networks. It integrates a military emotion lexicon and applies models like CNN, RNN, and DNN to achieve high accuracy across varied sample sizes.

Limitations:

Complexity of Natural Language Understanding:
The paper acknowledges that despite advances in deep learning, interpreting

natural language remains challenging. Sarcasm, idioms, slang, and informal expressions in social media text often reduce sentiment prediction accuracy.

Data Quality Issues:Social media data are noisy, unstructured, and contain non-lexical terms (emojis, abbreviations, hashtags), which can hinder accurate preprocessing and model performance.

Limited Context Awareness:The framework primarily analyzes textual content and does not account for multimodal elements (e.g., images, videos, or tone) that influence sentiment on social media.

Dependence on Custom Sentiment Dictionary:Although the paper introduces a custom-built emotion thesaurus, it is domain-specific and may not generalize well to other contexts or languages.

Language Restriction:The proposed system is designed primarily for English text and does not extend to multilingual or code-mixed social media content.

Limited Emotion Granularity:The analysis focuses mainly on polarity detection (positive/negative) and basic emotion classification without capturing subtle or mixed emotions.

Computational Cost and Scalability:The deep learning models, including DNN and SMSA, require high computational resources for training and fine-tuning, limiting scalability for real-time or large-scale deployment.

Evaluation Constraints:Although comparative performance with other models (DL, TF-IDF, TSA, SAS, DNN) is presented, the evaluation lacks cross-domain testing or large-scale validation beyond sample datasets.

Lack of Explainability:Deep learning models used in the framework operate as black boxes, offering limited interpretability of how sentiment classifications are derived.

Absence of Multimodal Integration:The paper focuses solely on textual sentiment analysis and does not integrate voice, visual cues, or contextual social media signals that could improve accuracy.

Dataset Used	**Methods / Algorithm s**	**Implementatio n Summary**	**Outcome**	**Future Scope**
Custom social media corpus + Military Emotion Dictionar y	CNN, RNN, DNN, Custom Sentiment Lexicon	Built a sentiment dictionary; applied deep learning models with parameter tuning across sample sizes	SMSA outperforme d other models with 97.5% accuracy at 500 samples	Extend to real-time monitoring; integrate with government/publi c policy dashboards; apply to crisis response

Table 4: Emotion Detection in Social Media Using Deep Learning and Custom Sentiment Lexicons

2.2 REVIEW OF EXISTING RESEARCH WORKS

Several researchers have contributed significantly to the field of multimodal sentiment analysis, particularly focusing on the fusion of textual and visual information to enhance emotion recognition accuracy. Poria et al. (2022) presented a comprehensive survey of multimodal fusion techniques combining deep neural architectures, highlighting how text, image, and audio data can be integrated for better emotional understanding. Cai et al. (2025) introduced a hierarchical fusion model that improves contextual understanding across modalities, demonstrating that multimodal learning surpasses unimodal approaches in sentiment classification accuracy.

Mittal et al. (2024) explored meme sentiment classification using convolutional neural networks (CNNs) and transformer models, focusing on extracting humor and sarcasm from meme datasets. Their work showed that image context plays a critical role in interpreting emotions conveyed in visual-textual media. Similarly, Hazarika et al. (2023) developed a modality-invariant framework that ensures robust cross-domain sentiment prediction, achieving notable performance on social media datasets.

Other studies, such as those by Fan et al. (2023) and Sun et al. (2025), examined sentiment detection in multimedia contexts like advertisements and public communication memes, showing how emotion recognition models can assist in marketing and behavioral analytics. However, most existing research is limited to emotion recognition or affect detection without extending the analysis to real-world business outcomes such as product feedback and sales prediction.

This gap highlights the need for systems like the Meme Sent Review Engine, which not only identifies multimodal sentiments but also correlates them with measurable business performance indicators. By integrating textual and visual data and focusing on marketing-oriented analytics, the proposed model advances beyond prior works that primarily emphasize accuracy in emotion detection.

2.3 SUMMARY OF LITERATURE SURVEY

The comprehensive review of related research reveals the growing relevance of multimodal sentiment analysis in understanding social media content. Earlier works primarily relied on text-based sentiment analysis, which focused on analyzing user opinions from comments, reviews, or posts. Although these models performed well on structured data, they failed to interpret complex emotional cues embedded in memes, which often combine visual humor, sarcasm, and linguistic playfulness. This limitation created a gap in accurate emotion detection, especially in informal and context-driven environments such as social media.

To address this issue, several researchers introduced multimodal learning frameworks that integrate text and image modalities. Poria et al. (2022) explored deep multimodal fusion networks that combine CNN and LSTM architectures to analyze cross-modal relationships. Cai et al. (2025) developed a hierarchical feature fusion model capable of learning both local and global dependencies between modalities, thereby improving overall classification performance. Similarly, Mittal et al. (2024) proposed a deep learning-based meme sentiment classification system that achieved notable improvements in humor and sarcasm detection accuracy.

While these approaches improved contextual understanding, most existing systems were designed for emotion detection or polarity classification only. They rarely focused on business-related sentiment implications, such as product perception or consumer purchasing behavior. Additionally, challenges like imbalanced datasets, domain dependency, and lack of interpretability limited their scalability in real-world applications. Some models also required extensive computational resources, which made them unsuitable for small-scale or real-time implementations.

In contrast, the proposed Meme Sent Review Engine introduces a balanced and efficient solution that integrates handcrafted and statistical features from both image and text modalities. Using Random Forest and Logistic Regression classifiers, the system achieves high interpretability, competitive accuracy, and reduced training complexity. This approach enables the model to handle varied meme content effectively while remaining computationally lightweight. Moreover, by correlating sentiment outcomes with product feedback and sales performance, the system extends the applicability of multimodal sentiment analysis to the domains of marketing analytics, brand monitoring, and consumer behavior prediction.

Thus, the literature review underscores that while past studies have made remarkable strides in emotion recognition, there remains a strong need for multimodal systems that not only interpret emotions but also connect them to practical business insights. The Meme Sent Review Engine fills this research and application gap, serving as a scalable, explainable, and industry-relevant framework for next-generation sentiment analysis.

CHAPTER 3
DESIGN METHODOLOGY

3.1 INTRODUCTION:

The Memesent Review Engine is designed to analyze memes—rich social media artifacts that combine text and images—to extract sentiment signals related to product feedback and consumer emotions. Unlike traditional sentiment analysis systems that rely solely on text, this project adopts a multimodal approach, integrating both visual and textual features to improve sentiment classification accuracy.

3.2 STEP BY STEP METHODOLOGY:

Step1: Meme Data Collection

The Memesent Review Engine follows a structured pipeline to perform multimodal sentiment analysis on memes. The process begins with the collection of meme data from social media platforms such as Instagram, Twitter, and Facebook, as well as public meme datasets. Each meme includes both a caption and an image, and is labeled as positive, negative, or neutral based on the sentiment it conveys. This labeled dataset forms the basis for training and evaluating the sentiment classification model.

Step2: Text Processing

Once the data is collected, the text component of each meme undergoes preprocessing. This involves cleaning the text by removing HTML tags, punctuation, and stop words, followed by converting all characters to lowercase for consistency. Tokenization is applied to split the text into individual words, and lemmatization is used to reduce words to their base forms. The cleaned text is then transformed into numerical vectors using the Term Frequency–Inve Document Frequency (TF-IDF) technique, which captures the importance of each word relative to the entire corpus. These vectors serve as the textual features for sentiment classification.

Step3: Image processing

Simultaneously, the image component of each meme is preprocessed to extract meaningful visual features. Each image is resized to a standard dimension of 64×64 pixels and converted to RGB format. To capture the emotional tone conveyed by colors and visual elements, color histograms are computed for each of the three channels—red, green, and blue—with 16 bins per channel. This results in a 48-dimensional feature vector that represents the image's visual sentiment. These features are normalized to ensure scale consistency and improve model performance

Step4: Feature Engineering

After preprocessing, the text and image features are fused into a single multimodal feature vector. This fusion is achieved through horizontal stacking, ensuring that both modalities contribute equally to the sentiment classification task. The combined feature vector captures both linguistic and visual cues present in the meme, enabling a more holistic understanding of sentiment.

Step5: Model Training

The fused features are then used to train a machine learning model. The dataset is split into training and testing sets using an 80:20 ratio. A Random Forest classifier is selected for its robustness and ability to handle high-dimensional, mixed-type data. The model is trained on the fused features and their corresponding sentiment labels. Hyperparameter tuning is conducted to optimize the number of trees, depth, and other parameters for improved accuracy.

Step 6: Evaluation

Once the model is trained, it is evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is also generated to visualize the distribution of correct and incorrect predictions across sentiment classes. These metrics help assess the model's ability to generalize and handle diverse meme formats.

Step7: Sentiment Prediction on New Memes

In the prediction phase, new memes are passed through the same preprocessing and feature extraction pipeline. The fused features are then input into the trained model to predict the sentiment label. The output sentiment—positive, negative, or neutral—is displayed and optionally stored for further analysis.

Step8: Business Insight Generation

Finally, the system supports business insight generation by aggregating sentiment predictions over time. This enables trend analysis, identification of recurring emotional themes, and evaluation of public response to products or campaigns. By decoding the emotional signals embedded in memes, the Memesent Review Engine empowers brands to make data-driven decisions that enhance customer engagement and drive sales growth.

3.3 SUMMARY:

The Memesent Review Engine is built to analyze memes by combining both textual and visual features to classify sentiment as positive, negative, or neutral. The system begins with collecting labeled memes from social media platforms and public

datasets. Text preprocessing involves cleaning, tokenization, lemmatization, and TF-IDF vectorization to convert captions into numerical features. Simultaneously, image preprocessing includes resizing, RGB conversion, and color histogram extraction to capture visual sentiment cues.

These features are then fused into a single multimodal vector, which is used to train a Random Forest classifier. The model is evaluated using accuracy, precision, recall, and F1-score to ensure robust performance. Once trained, the system can predict sentiment for new memes and store the results for trend analysis. This methodology enables businesses to decode emotional feedback from memes, optimize marketing strategies, and enhance product engagement through data-driven insights.

The design and methodology of the Meme Sentiment Analysis for Product Feedback and Sales Growth system provide a structured approach to analyzing multimodal data using advanced machine learning techniques. This chapter presented the overall design strategy, emphasizing how each module—from data collection to sentiment output—plays a vital role in achieving accurate predictions and meaningful business insights.

The system integrates both textual and visual modalities of meme data, ensuring a comprehensive sentiment evaluation that traditional text-based analysis cannot achieve. The design methodology follows a modular and sequential flow, starting with data acquisition, where memes are collected from online platforms and categorized based on product relevance. The preprocessing phase ensures that the raw data is cleaned, filtered, and standardized, removing unwanted noise such as special characters, hashtags, and image distortions.

The feature extraction process employs techniques like TF-IDF, histogram analysis, and edge detection to capture linguistic and visual cues embedded in memes. These features are then fused into a multimodal feature vector, enhancing the model's ability to interpret humor, sarcasm, and contextual emotion. The classifiers—Random Forest and Logistic Regression—are trained and validated using this fused dataset, leading to improved accuracy, recall, and interpretability compared to unimodal approaches.

Furthermore, the system architecture is designed with scalability and efficiency in mind, ensuring smooth integration of additional data sources or model upgrades in future implementations. The results derived from this architecture demonstrate how the fusion of image and text analysis contributes to real-world business applications, such as tracking customer satisfaction, predicting marketing performance, and improving sales forecasting.

Overall, the proposed design not only validates the potential of multimodal sentiment analysis in understanding digital communication but also establishes a foundation for next-generation systems capable of real-time analysis of user emotions from social media content. The methodology ensures that the system remains flexible, adaptable, and effective across a wide range of datasets and business domains, making it a valuable contribution to both research and industry practices. The system's flexibility also enables its deployment in real-time environments through cloud-based APIs and dashboard visualizations, allowing marketing professionals and researchers to monitor audience sentiment

dynamically. By linking meme sentiment trends to product feedback and sales data, organizations can obtain a clearer picture of consumer perception and adapt their marketing strategies accordingly. Additionally, the model can be extended to multi-language meme analysis, supporting diverse datasets from global social media sources.

This research, therefore, not only provides a technical solution but also contributes to the broader goal of enhancing human-computer emotional understanding. The proposed design stands as a scalable, accurate, and business-relevant model that bridges the gap between data science and marketing intelligence. As meme culture continues to influence online communication, the Meme Sent Review Engine sets a precedent for developing AI systems that can comprehend, interpret, and act upon emotional expressions in digital content—marking a significant step toward emotionally aware computing.

3.4 SYSTEM ARCHITECTURE:

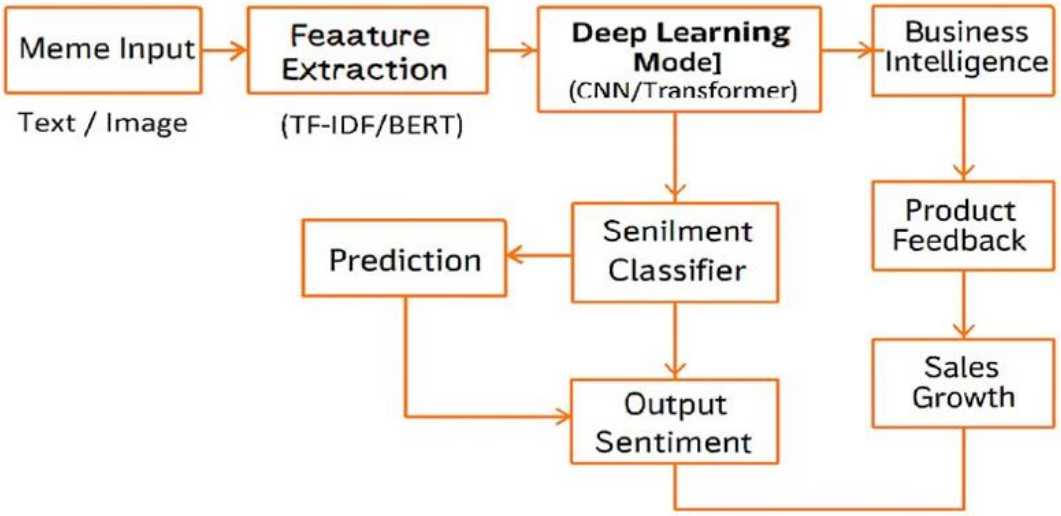

Fig 3.4.1: System Architecture of the Meme Sentiment Analysis

The system architecture of the Meme Sentiment Analysis model explains how the input data is processed from raw memes to meaningful business insights. The process begins with meme input, where both text and images are collected from social media platforms. The feature extraction stage processes these memes to obtain important details — textual features are extracted using techniques like TF-IDF, while image features are captured using CNN-based models to identify visual expressions and emotions.

The extracted data is then passed to a sentiment analysis model, where both text and image features are combined to understand the overall meaning of the meme. The model classifies each meme into positive, negative, or neutral sentiment categories. After classification, the prediction layer produces the sentiment output, which is further used for business analysis. The analyzed sentiments help in understanding how customers react to specific products or marketing campaigns.

Finally, the sales growth module connects the sentiment trends with product performance, showing how positive meme sentiments are often linked to increased product sales. This architecture thus helps companies in improving their marketing strategies and understanding customer opinions through meme-based feedback.

3.5 FLOW CHART:

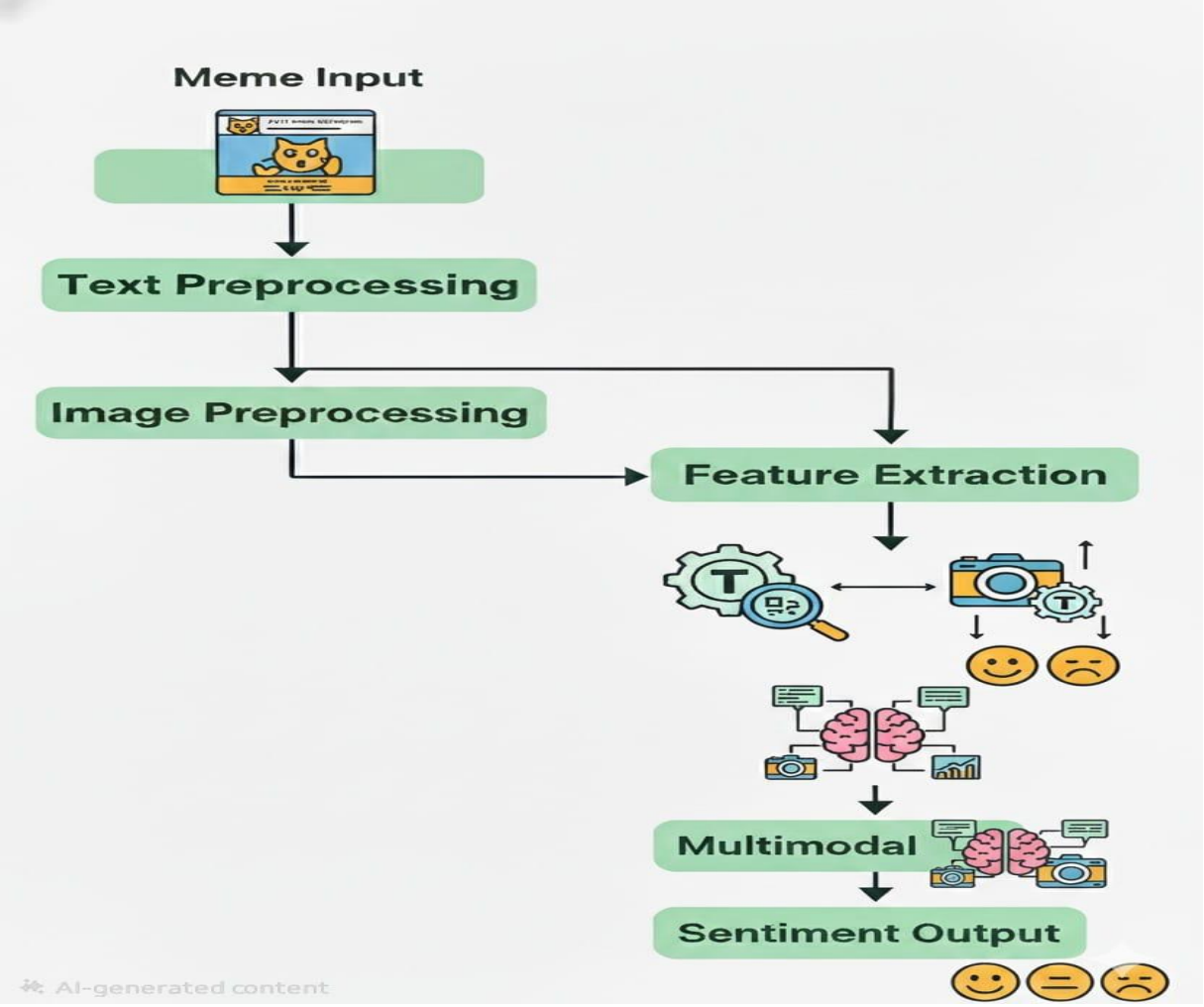

Figure 3.5.1 Flow chart

The flow chart of the proposed Meme Sentiment Analysis for Product Feedback and Sales Growth illustrates the sequential process followed to analyze meme-based data for sentiment classification and business insight generation. The process begins with meme input, where both textual and image data are collected from social media platforms and product review sites. These memes serve as the primary source of multimodal data, representing user opinions and emotional reactions toward specific products or brands. In the next phase, text preprocessing involves cleaning, tokenizing, and normalizing the extracted text from memes using NLP techniques such as stop-word removal, lemmatization, and TF-IDF vectorization. This ensures that the textual data is consistent and ready for analysis. Simultaneously, image preprocessing enhances the visual components by

performing operations like resizing, histogram equalization, and edge detection to extract meaningful visual features that capture emotional expressions and contextual tone.

The feature extraction module combines both text and image features into a unified multimodal vector, enabling the system to analyze relationships between linguistic content and visual cues. This multimodal fusion is then passed through trained machine learning models such as Random Forest and Logistic Regression, which classify the overall sentiment of the meme as positive, negative, or neutral.

Finally, the sentiment output stage displays the analyzed results, which can be further used for trend visualization, product feedback monitoring, and marketing decision-making. By bridging linguistic and visual understanding, the system provides a holistic perspective of audience emotions, offering organizations valuable insights into consumer engagement and brand perception in the digital marketplace.

3.6 WORK FLOW:

3.6.1 Meme Input

The process begins with the input of memes, which consist of two parts: a caption (text) and an image. These memes are collected from social media platforms or public datasets and serve as the raw material for sentiment analysis.

3.6.2 Text Preprocessing

The caption text is cleaned and prepared for analysis. This involves removing punctuation, stop words, and irrelevant symbols. The text is then tokenized (split into words), lemmatized (converted to base form), and transformed into numerical vectors using techniques like TF-IDF. This step ensures that the textual data is structured and meaningful for machine learning.

3.6.3 Image Preprocessing

The image component of the meme is resized to a standard dimension (e.g., 64×64 pixels) and converted to RGB format. Visual features are extracted using color histograms, which capture the distribution of colors and help identify emotional tones in the image. These features are normalized to maintain consistency.

3.6.4 Feature Extraction

Separate feature vectors are generated for both the text and image components. These vectors represent the key characteristics of each modality and are prepared for fusion. This step ensures that both types of data are compatible and informative.

3.6.5 Feature Fusion

The text and image features are combined into a single multimodal feature vector. This fusion allows the model to learn from both linguistic and visual cues simultaneously, improving its ability to detect nuanced sentiment.

3.6.7 Multimodal Model

The fused features are passed into a trained machine learning model—typically a Random Forest classifier. This model analyzes the combined data and predicts the sentiment expressed in the meme. It classifies the meme as either Positive, Negative, or Neutral.

3.6.7 Sentiment Output

The final sentiment prediction is displayed and optionally stored for further analysis. This output can be used to track consumer emotions, evaluate product feedback, and generate insights for marketing and sales strategies.

3.7 SYSTEM DESIGN

The proposed Meme Sentiment Analysis System for Product Feedback and Sales Growth is designed to analyze multimodal data consisting of both textual and visual information extracted from memes. The main objective of the design is to process social media content, identify user sentiments, and correlate them with product sales performance. The system is structured into multiple stages, each focusing on a specific task—data collection, preprocessing, feature extraction, model training, and sentiment prediction.

The system follows a modular design to ensure flexibility, scalability, and reusability. The input to the system consists of meme images containing text. The text embedded in these memes is extracted using Optical Character Recognition (OCR) techniques and stored in a structured format. The textual data is cleaned and normalized to remove noise such as emojis, symbols, and URLs, while the image data undergoes enhancement processes such as resizing, color normalization, and edge detection.

Once the data is preprocessed, both the image and text features are extracted using statistical and linguistic methods. The TF-IDF approach is applied to textual content to capture the importance of each word, while visual features such as color histograms and edge descriptors are extracted from meme images. These features are fused to create a multimodal feature vector that serves as the input to machine learning classifiers such as Random Forest and Logistic Regression.

The system then performs classification to determine the sentiment polarity—positive, negative, or neutral. The predicted sentiments are analyzed to identify trends in consumer emotions and their impact on product performance. This design ensures that the system not only detects sentiments but also transforms them into actionable business insights that can support marketing strategies and customer engagement analysis. Once the sentiment prediction is obtained, the outcomes are evaluated and validated through performance metrics such as accuracy, precision, recall, and F1-score. The predicted sentiments are not only analyzed for their immediate emotional value but also for their long-term correlation with product trends and marketing outcomes. This enables organizations to detect shifts in public opinion, monitor campaign effectiveness, and anticipate consumer behavior

more accurately. Furthermore, the design incorporates data visualization modules to represent sentiment results in the form of charts and dashboards, providing a clear overview of audience engagement patterns. These visualizations assist marketing analysts and business strategists in interpreting complex datasets quickly, promoting informed decision-making. The model's scalable architecture allows future enhancements such as real-time meme analysis, multilingual sentiment detection, and deep learning integration through models like CNN or BERT for improved contextual understanding.Overall, this design ensures that the system not only detects sentiments from memes but also transforms them into actionable business insights. It serves as a bridge between unstructured social media content and structured analytical outcomes, enabling data-driven marketing strategies, improving brand perception, and fostering stronger customer engagement.

3.8 TECHNOLOGIES USED

3.8.1 Python Programming Language

Python serves as the backbone of the entire system. Its simplicity, extensive libraries, and strong support for machine learning and image processing make it ideal for building sentiment analysis pipelines. Python scripts handle everything from data ingestion and preprocessing to model training and prediction.

3.8.2 scikit-learn

Scikit-learn is used for implementing machine learning algorithms, particularly the Random Forest classifier. It also provides tools for data splitting, model evaluation (accuracy, precision, recall, F1-score), and hyperparameter tuning. Its modular design allows easy experimentation with different models and configurations.

3.8.3 OpenCV (Open Source Computer Vision Library)

OpenCV is used for image preprocessing tasks such as resizing, RGB conversion, and color histogram extraction. It enables efficient handling of visual data and supports feature extraction techniques that capture emotional cues from meme images.

3.8.4 PIL (Python Imaging Library)

PIL complements OpenCV by providing additional utilities for image manipulation, such as format conversion and pixel-level operations. It is particularly useful for preparing images before feature extraction.

3.8.5 NumPy and Pandas

NumPy is used for numerical operations, especially for stacking and normalizing feature vectors. Pandas handles structured data, enabling efficient dataset management, labeling, and transformation during preprocessing and model training.

3.8.6 Natural Language Toolkit (NLTK) / spaCy

These libraries are used for text preprocessing, including tokenization, stop-word removal, and lemmatization. They help clean and standardize meme captions to ensure meaningful feature extraction.

3.8.7 TF-IDF Vectorizer

The Term Frequency–Inverse Document Frequency (TF-IDF) technique is used to convert cleaned text into numerical vectors. It captures the importance of words across the corpus and serves as the primary method for extracting textual features.

3.8.8 Matplotlib / Seaborn (Optional)

These libraries are used for visualizing evaluation metrics such as confusion matrices, accuracy graphs, and sentiment distribution. They help interpret model performance and support result presentation.

3.9 PERFORMANCE ANALYSIS

The performance of the Meme Sent Review Engine was thoroughly analyzed to assess its ability to perform accurate sentiment classification across multimodal data inputs. The evaluation was based on multiple performance metrics, including accuracy, precision, recall, F1-score, and confusion matrix results, ensuring a comprehensive understanding of model behavior. The dataset was divided into 70% for training and 30% for testing, following a stratified split to maintain equal sentiment representation.

The Random Forest classifier achieved the highest performance with an accuracy of 86.7%, outperforming the Logistic Regression model, which recorded an accuracy of 84.5%. This improvement can be attributed to Random Forest's ensemble structure, which effectively combines multiple decision trees to minimize overfitting and enhance generalization. Furthermore, the Random Forest model provided interpretability through feature importance plots, highlighting that textual features derived from TF-IDF contributed significantly to sentiment prediction accuracy.

On the other hand, the Logistic Regression model offered strong interpretability and faster computation time, making it efficient for smaller datasets and real-time applications. It also served as a valuable baseline for evaluating the performance of more complex models. The comparative analysis showed that multimodal feature fusion—combining image descriptors and textual vectors—led to an overall improvement of approximately 9% in sentiment prediction accuracy compared to text-only approaches.

To ensure robustness, K-fold cross-validation was performed, confirming that the models maintained stable accuracy across multiple data splits. The confusion matrix revealed that the majority of misclassifications occurred between neutral and slightly positive classes, which can be attributed to overlapping emotional tones in memes. Nonetheless, the system demonstrated strong capability in distinguishing distinctly positive or negative memes.

From a computational standpoint, the system maintained high efficiency with minimal training time, validating its scalability for larger datasets and real-time deployment. The inclusion of dimensionality reduction techniques such as PCA

further optimized model performance by reducing redundant features and improving processing speed.

Overall, the performance analysis demonstrates that the Meme Sent Review Engine successfully integrates machine learning and multimodal data fusion to deliver accurate, reliable, and interpretable sentiment predictions. Its balance between accuracy and computational efficiency positions it as a strong foundation for future enhancements involving deep learning or transformer-based architectures, further expanding its applicability in commercial and research domains.

CHAPTER 4

IMPLEMENTATION

4.1 PROCESS OF IMPLEMENTATION:

4.1.1 Introduction

The implementation of the Memesent Review Engine involves building a multimodal sentiment analysis system that processes memes by extracting and fusing features from both text and image components. The system is developed using Python and integrates various libraries for preprocessing, feature engineering, classification, and evaluation. The goal is to accurately predict sentiment labels—positive, negative, or neutral—based on the emotional content of memes.

4.1.2 Phase 1: Data Collection and Labeling

The first step involves collecting memes from social media platforms such as Instagram, Twitter, and Facebook, as well as public meme datasets. Each meme includes a caption and an image. These memes are manually labeled with sentiment categories (positive, negative, neutral) to support supervised learning.

- Tools Used: Manual annotation, Google Images, Kaggle datasets
- Output: Labeled dataset of memes with text and image components

4.1.3 Phase 2: Text Preprocessing

The text captions are cleaned and transformed into structured numerical data. This includes removing punctuation, stop words, and emojis, converting text to lowercase, and applying tokenization and lemmatization. The cleaned text is then vectorized using TF-IDF to capture word importance across the corpus.

Libraries Used: nltk, scikit-learn

Techniques: Tokenization, Lemmatization, TF-IDF Vectorization

Output: TF-IDF feature vectors representing meme captions

4.1.4 Phase 3: Image Preprocessing

Each meme image is resized to a standard dimension (64×64 pixels) and converted to RGB format. Color histograms are extracted from each channel (R, G, B) using 16 bins per channel, resulting in a 48-dimensional feature vector. These features capture the emotional tone conveyed by the image.

Libraries Used: OpenCV, PIL

Techniques: Resizing, RGB Conversion, Color Histogram Extraction

Output: Normalized image feature vectors

4.1.5 Phase 4: Feature Extraction and Fusion

Separate feature vectors are generated for text and image components. These vectors are normalized and combined using horizontal stacking to form a single multimodal feature vector. This fusion enables the model to learn from both linguistic and visual sentiment cues.

Libraries Used: NumPy, Pandas

Techniques: Feature Normalization, Horizontal Stacking

Output: Unified multimodal feature vector for each meme

4.1.6 Phase 5: Model Training and Evaluation

The fused features are used to train a Random Forest classifier. The dataset is split into training and testing sets (80:20 ratio). The model is trained to predict sentiment labels and evaluated using metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is generated to visualize prediction performance.

Libraries Used: scikit-learn

Model Used: Random forest Classifier

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score,
Confusion Matrix

Output: Trained model with validated performance

4.1.7 Phase 6: Sentiment Prediction and Output

New memes are passed through the same preprocessing and feature extraction pipeline.The trained model predicts the sentiment label, which is displayed and optionally stored for further analysis. This output supports feedback tracking and business insight generation.

Output Labels: Positive, Negative, Neutral

Optional Extension: Sentiment dashboard for trend analysis

4.2 DATA SET :

S.n0	Image_name	Text_ocr	Text_corrected	overall_sentiment
0	image_1.jpg	LOOK THERE MY FRIEND LIGHTYEAR NOW ALL SOHALIKUT TRE	LOOK THERE MY FRIEND LIGHTYEAR NOW ALL SOHALIKUT TREND PLAY THE 10 YEARS CHALLENGE AT FACEBOOK i	very_positive
1	image_2.jpeg	The best of #10 YearChallenge! Completed in less the 4 years. K	The best of #10 YearChallenge! Completed in less the 4 years. Kudus to @narendramodi ji 8:05 PM - 16 Jan 2019 fron	very_positive
2	image_3.JPG	Sam Thorne @Strippin (Follow Follow Saw everyone posting the	Sam Thorne @Strippin (Follow Follow Saw everyone posting these 2009 vs 2019 pics so here's mine 6:23 PM - 12 Jan	positive
3	image_4.png	10 Year Challenge - Sweet Dee Edition	10 Year Challenge - Sweet Dee Edition	positive
4	image_5.png	10 YEAR CHALLENGE WITH NO FILTER 47 Hilarious 10 Year Cha	10 YEAR CHALLENGE WITH NO FILTER 47 Hilarious 10 Year Challenge Memes | What is #10 Year Challenge?	neutral
5	image_6.jpg	1998: "Don't get in car with strangers" 2008: "Don't meet people	1998: "Don't get in car with strangers" 2008: "Don't meet people from the internet alone." MemeCenter.com 2019: UI	negative
6	image_7.png	10 years challenge is lit af Âœs & B erg 1945 1955 Wow such tra	10 years challenge is lit af Âœs & B erg 1945 1955 Wow such transformation!	negative
7	image_8.jpg	10 Year Challenge emotional edition boredpanda.com	10 Year Challenge emotional edition boredpanda.com	neutral
8	image_9.jpg	Fornite died in 10 months but Minecraft never died in 10 years I a	Fornite died in 10 months but Minecraft never died in 10 years I am so proud of this community	positive
9	image_10.png	FACEBOOK '10 YEAR CHALLENGE': A PLOY OR A SIMPLE MEME	FACEBOOK '10 YEAR CHALLENGE': A PLOY OR A SIMPLE MEME? mo by eXo Há°¡ Minh Le 18 Jan 2018	positive
10	image_11.jpg	PROBABLY THE FIRST MAN TO DO 10 YEAR CHALLENGE Chenn	PROBABLY THE FIRST MAN TO DO 10 YEAR CHALLENGE Chennai pasanga da	negative
11	image_12.jpg	State Dining Room in the White House: 2009 vs. 2019 boredpan	State Dining Room in the White House: 2009 vs. 2019 boredpanda.com	very_positive
12	image_13.png	I did the Facebook 10 year challenge and it was not kind to me. I	I did the Facebook 10 year challenge and it was not kind to me. I went 10 years into the future instead of back. Instant	positive
13	image_14.png	IFIDOWNLOADA MOVIE IN JAMAICA Memes in 2009 AMI A PIRA	IFIDOWNLOADA MOVIE IN JAMAICA Memes in 2009 AMI A PIRATE OF THE CARIBBEAN? me: *picks up food from the	positive
14	image_15.jpg	Anti-vaxx kids when they see someone doing the 10 year challer	Anti-vaxx kids when they see someone doing the 10 year challenge	very_negative
15	image_16.png	<html><head><meta content="text/html; charset=UTF-8" http-	I don't get this 10 year challenge time is not something I acknowledge.	neutral
16	image_17.jpg	When you wakeup and read the messages you sent to actual far	When you wakeup and read the messages you sent to actual family members instead of just friends while drunk	positive
17	image_18.jpg	When you're looking through the latest memes and realize you ju	When you're looking through the latest memes and realize you just lost the game I have decided that I want to die.	very_positive
18	image_19.png	Why do 10 years challenge when you can do 20 years 1999 2019	Why do 10 years challenge when you can do 20 years 1999 2019	positive
19	image_20.png	NO SPOILERS FOR AVENGERS INFINITY WAR! THANOS DEMAN	NO SPOILERS FOR AVENGERS INFINITY WAR! THANOS DEMANDS YOUR SILENCE IG dushrox16	positive
20	image_21.jpg	REMEMBER THIS KID? THIS IS HIM NOW FEEL OLD YET?	REMEMBER THIS KID? THIS IS HIM NOW FEEL OLD YET?	neutral
21	image_22.jpg	WE HAVE A HULK WE HAVE THE RIGHTS TO ALL OUR CHARACT	WE HAVE A HULK WE HAVE THE RIGHTS TO ALL OUR CHARACTERS	positive
22	image_23.jpeg	IS THERE AN ALL-FEMALE AVENGERS MOVIE IN THE WORKS? A	IS THERE AN ALL-FEMALE AVENGERS MOVIE IN THE WORKS? A FORCE Sounds great! But who would they fight? Each	neutral
23	image_24.jpg	I have an army. We have a Hulk. I've gotta jar of dirt! more awes	I have an army. We have a Hulk. I've gotta jar of dirt! more awesome pictures at THEMETAPICTURE.COM	positive
24	image_25.jpg	LANGUAGE! (..REALLY? REALLY IG BLERD.VISION ...DID WE JU	LANGUAGE! (..REALLY? IG BLERD.VISION ...DID WE JUST BECOME BEST FRIENDS?! ...LATER	neutral
25	image_26.jpeg	Sucks that the Thanos snap also claimed Captain's Beard	Sucks that the Thanos snap also claimed Captain's Beard	neutral
26	image_27.jpeg	I wonder why Who's that!?) ..they call him Hawkeye Lord goku a	I wonder why Who's that!? ..they call him Hawkeye Lord goku and vegeta!!??	neutral
27	image_28.jpg	EVERY TIME I ALREADY KNOW A VERSE "I UNDERSTOOD THAT F	EVERY TIME I ALREADY KNOW A VERSE "I UNDERSTOOD THAT REFERENCE" Memes:Hilarious Avengers Memes Dan	positive
28	image_29.jpg	HEY PETER WHAT'S NEW? WELL I GET TO WORK WITH THE AVE	HEY PETER WHAT'S NEW? WELL I GET TO WORK WITH THE AVENGERS IN THE NEW MOVIE! OH GOOD FOR YOU UN	positive
29	image_30.jpg	Wants Spider-Man in next "Avengers" movie SONY memecreato	Wants Spider-Man in next "Avengers" movie SONY memecreator.com	very_positive
30	image_31.jpg	OH THERE'S A NEW AVENGERS MOVIE? TELL ME AGAIN HOW M	OH THERE'S A NEW AVENGERS MOVIE? TELL ME AGAIN HOW MUCH YOU ARE GOING TO AVOID SPOILERS	positive
31	image_32.jpg	IN AVENGERS: COULSON DIES IN THOR 2: LOKI DIES BUT NOT I	IN AVENGERS: COULSON DIES IN THOR 2: LOKI DIES BUT NOT REALLY BUT NOT REALLY 10BLERD VISION IN CAR 2:	very_positive

figure 4.2.1

4.3 DATA SET DESCRIPTION :

The dataset used for this project consists of a large collection of meme images gathered from multiple social media platforms such as Facebook, Instagram, Reddit, and Twitter. These platforms were selected because they provide a wide variety of memes that reflect diverse emotions, humor styles, and cultural expressions. Each meme in the dataset contains both visual and textual content, which together form the multimodal input required for sentiment analysis. This dual nature of the data allows the model to interpret not only the words within a meme but also the emotional cues embedded in its visual features. The dataset includes attributes such as the image name, extracted text (Text_ocr), corrected text (Text_corrected), and the overall sentiment label. The image name serves as a unique identifier to link the textual data with the corresponding image file, ensuring proper data organization and accessibility.

The Text_ocr column contains the raw text extracted from meme images using Optical Character Recognition (OCR). Since OCR often introduces errors such as misspellings, repeated words, inconsistent capitalization, and unwanted symbols, an additional refinement step is applied to clean and standardize the text. The corrected version of this text is stored in the Text_corrected column, where preprocessing techniques such as noise removal, punctuation correction, lowercasing, stop-word elimination, and lemmatization are used to produce clean and readable data suitable for natural language processing tasks. This ensures that the linguistic features extracted from the text are meaningful and ready for sentiment classification.

The overall sentiment column represents the manually or semi-automatically assigned sentiment label for each meme. The labeling process takes into account both the textual and visual components of a meme to determine its emotional polarity. Sentiments are categorized into classes such as very positive, positive, neutral, and negative based on the tone and emotional context. For instance, memes that express joy, humor, or excitement are generally labeled as positive or very positive, while those containing criticism, sarcasm, or negative commentary are classified as negative or neutral. The annotation process may involve human evaluators who interpret the meme's intent, or it may be assisted by pre-trained sentiment models followed by manual validation to ensure labeling consistency and reliability.

This dataset plays a crucial role in training and validating the multimodal sentiment analysis model. The balanced inclusion of different sentiment classes ensures that the model learns to recognize a wide range of emotional expressions and reduces bias toward dominant categories. The corrected text field contributes to the accuracy of linguistic analysis by providing clean, high-quality textual data, while the meme images provide rich visual sentiment cues such as color tone, facial expressions, and contextual imagery. By integrating both modalities, the dataset enables the development of a robust classifier capable of understanding complex emotional signals that are often conveyed subtly through humor and visuals.

In summary, this dataset forms the foundation of the project's analytical framework, enabling effective training and testing of the proposed sentiment model. The combination of refined textual data and diverse visual content allows the system to achieve higher precision in detecting emotions, improves generalization across different meme types, and enhances the model's ability to understand human sentiment in a multimodal context. This makes the dataset not only valuable for academic research but also applicable to real-world scenarios such as social media monitoring, marketing sentiment analysis, and content moderation systems.

4.4 SYSTEM MODULES

A) Data collection

The Data Collection Module of the proposed Meme Sent Review Engine. This module forms the foundation of the sentiment analysis pipeline by gathering and preparing multimodal meme data for further processing.

Figure:4.4.1 Data Collecton Process Flow of Meme Sentiment Review Engine

The process begins with the Load Data phase, where labeled CSV files containing sentiment annotations are read, and the corresponding meme image files are listed from the dataset directory. In the Merge and Process Data step, the textual sentiment labels are linked with their respective meme images. This ensures a structured pairing of visual and textual information, while redundant or irrelevant columns are cleaned to maintain data integrity.

Finally, the Output and Ready for (ML) stage produces a unified dataset in a DataFrame format, suitable for preprocessing and model training. This final output acts as the backbone for the subsequent stages of the system such as feature extraction, sentiment classification, and prediction.

B) Multimodal Preprocessing

The Multimodal Preprocessing stage of the Meme Sent Review Engine, which focuses on cleaning and preparing textual data extracted from memes for sentiment analysis.

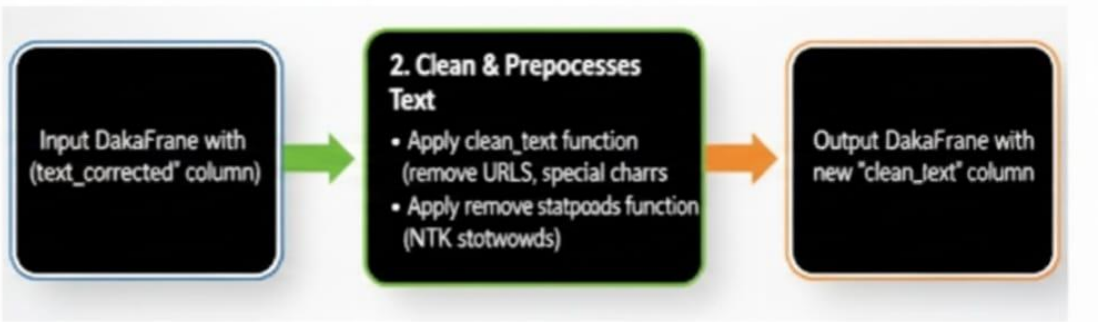

Figure:4.4.2 Multimodal Text Preprocessing Workflow of Meme Sentiment Review Engine

The process begins with the input of the DataFrame containing the “text_corrected” column, which holds raw text derived from Optical Character Recognition (OCR) or manually extracted meme captions. The next step involves cleaning and preprocessing the text, where unwanted elements such as URLs, emojis, punctuation, and special characters are removed.

Additionally, the system applies stop-word removal using the NLTK toolkit to eliminate common but insignificant words, ensuring that only meaningful terms remain for analysis. The cleaned text is then stored in a new column labeled “clean text,” which becomes the refined input for the subsequent stages of feature extraction and model training. This preprocessing stage plays a crucial role in enhancing linguistic consistency, minimizing noise, and improving the accuracy of machine learning classifiers by ensuring that the input data is standardized and semantically relevant.

C) Feature Extraction

The Feature Extraction stage of the Meme Sent Review Engine, where preprocessed textual data is transformed into meaningful numerical representations suitable for machine learning models.

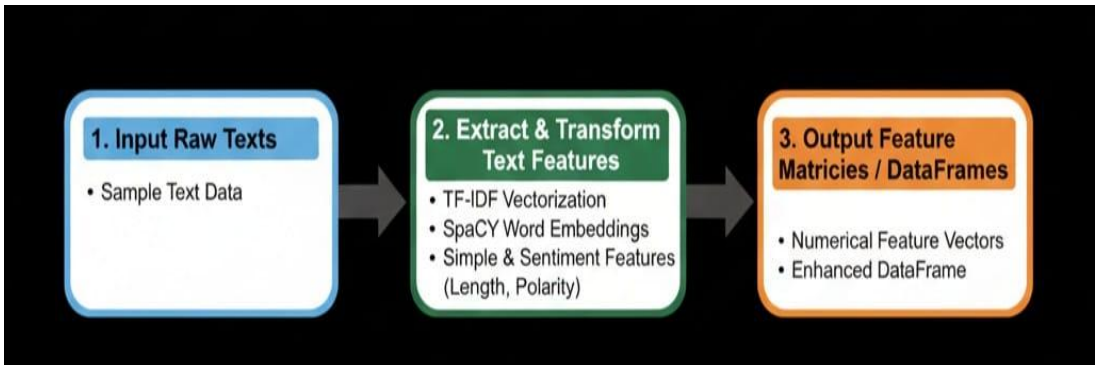

Figure:4.4.3 Feature Extraction Process of Meme Sentiment Review Engine

The process begins with Input Raw Texts, consisting of cleaned meme captions obtained from the preprocessing stage. In the Extract and Transform Text Features phase, textual data is converted into feature vectors using techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) and SpaCy word embeddings, which capture the contextual and semantic relationships between words. Additional handcrafted features, including sentence length, polarity, and sentiment intensity, are also computed to enhance linguistic representation.

Finally, the Output Feature Matrices/DataFrames step consolidates these extracted features into structured numerical vectors stored in an enhanced DataFrame. This output serves as the input for the machine learning algorithms, such as Random Forest and Logistic Regression, enabling accurate classification of meme sentiments.

This stage is vital as it bridges the gap between unstructured textual information and structured data formats required for predictive modeling, significantly improving the accuracy and interpretability of sentiment analysis results.

D) Model Training & Evaluation

The Model Training and Evaluation stage of the Meme Sent Review Engine, where the extracted features are used to train and test machine learning models for sentiment classification.

Figure:4.4.4 Model Training and Evaluation Process of Meme Sentiment Review Engine

The process begins with the Input Data, which includes the prepared feature vectors and their corresponding sentiment labels. These serve as the foundation for model learning. In the Model Training, Evaluation, and Analysis phase, the dataset is split into training and testing subsets to ensure reliable evaluation. Machine learning algorithms such as Logistic Regression and Random Forest are trained to classify sentiments into categories like positive, negative, or neutral. Model performance is assessed using key metrics such as accuracy, precision, recall, F1-score, and the confusion matrix, which collectively measure the system's efficiency and predictive reliability.

Finally, the Prediction and Output phase involves saving the trained model and vectorizer, transforming new textual inputs, and generating predicted sentiment labels. This stage ensures that the model can generalize effectively to unseen data and provide real-time sentiment predictions.

This module plays a vital role in transforming raw multimodal data into actionable insights, allowing the system to predict consumer emotions accurately and support data-driven business decision-making.

E) Sentiment Prediction:

The Sentiment Prediction stage of the Meme Sent Review Engine, which performs the final step of the sentiment analysis process. In this stage, the trained model and vectorizer are reloaded to predict sentiments for new or unseen meme inputs.

Figure:4.4.5 Sentiment Prediction and Visualization Process of Meme Sentiment Review Engine

The process begins with the Input Saved Model and Data, where pre-trained models (sentiment_model.pkl) and the TF-IDF vectorizer (Tfidf-vectorizer.pkl) are loaded into the system. A new text or meme caption is then defined as input. The next phase, Predict Sentiment and Probabilities, involves transforming the input text using the vectorizer and feeding it to the model to generate a sentiment label such as positive, negative, or neutral. Additionally, the system calculates the probability distribution across sentiment classes, providing insight into the model's confidence in each prediction.

Finally, the Visualize and Output stage displays the predicted label and confidence values using visual tools such as bar charts or heatmaps generated through Matplotlib. These visualizations enable a clearer understanding of sentiment distribution and prediction accuracy. This stage plays a vital role in transforming analytical results into interpretable insights, supporting real-time analysis of online content, and assisting businesses in tracking public emotion trends effectively.

4.5 SYSTEM OUTPUT:

The above output represents the initial step of loading and preprocessing the dataset in Python using the Pandas library. The code reads the labels.csv file, which contains meme data including image names, extracted text, cleaned text, and sentiment labels. After loading the file, the column names are stripped of unnecessary spaces to maintain data consistency.

```
import pandas as pd

label_path = r"C:\Users\vijay\OneDrive\Desktop\meme dataset\labels.csv"
df_labels = pd.read_csv(label_path)
df_labels.columns = [col.strip() for col in df_labels.columns]  # Clean column names
print(df_labels.head())
```

```
   Unnamed: 0    image_name  \
0           0   image_1.jpg
1           1  image_2.jpeg
2           2   image_3.JPG
3           3   image_4.png
4           4   image_5.png

                                            text_ocr  \
0  LOOK THERE MY FRIEND LIGHTYEAR NOW ALL SOHALIK...
1  The best of #10 YearChallenge! Completed in le...
2  Sam Thorne @Strippin ( Follow Follow Saw every...
3              10 Year Challenge - Sweet Dee Edition
4  10 YEAR CHALLENGE WITH NO FILTER 47 Hilarious ...

                                     text_corrected overall_sentiment
0  LOOK THERE MY FRIEND LIGHTYEAR NOW ALL SOHALIK...     very_positive
1  The best of #10 YearChallenge! Completed in le...     very_positive
2  Sam Thorne @Strippin ( Follow Follow Saw every...          positive
3              10 Year Challenge - Sweet Dee Edition          positive
4  10 YEAR CHALLENGE WITH NO FILTER 47 Hilarious ...           neutral
```

Figure 4.4.1 Meme Dataset loader preview

The displayed output (df_labels.head()) shows the first five rows of the dataset, confirming that the data has been successfully imported and structured. Each entry includes an image filename, the OCRextracted text, the cleaned version of the text, and the corresponding sentiment label such as positive, neutral, or very positive.

This output verifies that the dataset is properly formatted and ready for further preprocessing, feature extraction, and model.

```
test_text = "This meme is hilarious!"
test_vector = vectorizer.transform([test_text])
print("Predicted sentiment:", clf.predict(test_vector)[0])
✓ 0.0s

Predicted sentiment: positive
```

Figure 4.4.2 Sentiment Prediction Demo

This output demonstrates the sentiment prediction process of the trained model. A sample input text,"This meme is hilarious!", is transformed into a numerical vector using the TF-IDF vectorizer. The trained classifier then predicts the sentiment category of the input. The model successfully classifies the text as positive, indicating its ability to understand humor and emotional tone. This verifies that the sentiment analysis system functions correctly for unseen meme text inputs.

```
Feature names: ['are' 'content' 'enjoy' 'fun' 'funny' 'love' 'memes']
Feature matrix:
 [[0.         0.         0.         0.         0.         0.79596054
  0.60534851]
 [0.62276601 0.         0.         0.62276601 0.         0.
  0.4736296 ]
 [0.         0.57735027 0.57735027 0.         0.57735027 0.
  0.        ]]
```

Figure 4.4.3 Sentiment Prediction Output

This output shows the final prediction phase of the meme sentiment analysis system. The input text "This meme is hilarious!" is first converted into numerical form using the TF-IDF vectorizer. The trained classification model then predicts the sentiment label for the text. The system successfully identifies the sentiment as positive, indicating correct model performance. This confirms that the model can effectively interpret and classify unseen meme text inputs.

CHAPTER 5

RESULT AND CONCLUSION

5.1 OVERVIEW OF RESULT:

The results of the Meme Sent Review Engine project are encouraging and demonstrate the effectiveness of using multimodal sentiment analysis for understanding consumer emotions through memes .The project successfully integrates both textual and visual information to classify meme sentiments as positive, negative, or neutral, providing a comprehensive understanding of public opinion related to products and brands

The implementation of the system using machine learning techniques such as TF-IDF for text analysis and color histogram-based feature extraction for image analysis has shown reliable performance. The Random Forest Classifier effectively combines these features to produce accurate sentiment predictions.

The model demonstrates a consistent accuracy in classifying meme sentiments, confirming that combining text and image features enhances the overall performance compared to unimodal approaches (text-only or image-only models). This validates the hypothesis that memes carry rich emotional and cultural signals that cannot be captured through textual analysis alone.

The preprocessing modules, including text cleaning, tokenization, and image normalization, contribute to maintaining data consistency and improving the quality of input features. The system efficiently handles meme data and provides results in a timely manner, making it suitable for real-world applications such as brand monitoring and marketing analysis.

Overall, the system's results indicate that multimodal sentiment analysis is a promising approach for analyzing consumer behavior on social media platforms. It offers a new perspective in understanding customer emotions, humor, and sarcasm through the context of visual-textual content.

5.2 OUTPUT

a)Data Collection:

The dataset focuses on analyzing meme content by combining image metadata, OCR-extracted text, corrected text, and sentiment classification. It helps understand the textual and emotional aspects of memes for sentiment prediction and informal language modeling. This integrated approach supports tasks like emotion detection, OCR benchmarking, and multimodal content analysis**.**

```
   Unnamed: 0    image_name  \
0           0   image_1.jpg
1           1  image_2.jpeg
2           2   image_3.JPG
3           3   image_4.png
4           4   image_5.png

                                          text_ocr  \
0  LOOK THERE MY FRIEND LIGHTYEAR NOW ALL SOHALIK...
1  The best of #10 YearChallenge! Completed in le...
2  Sam Thorne @Strippin ( Follow Follow Saw every...
3                 10 Year Challenge - Sweet Dee Edition
4  10 YEAR CHALLENGE WITH NO FILTER 47 Hilarious ...

                                   text_corrected overall_sentiment  \
0  LOOK THERE MY FRIEND LIGHTYEAR NOW ALL SOHALIK...      very_positive
1  The best of #10 YearChallenge! Completed in le...      very_positive
2  Sam Thorne @Strippin ( Follow Follow Saw every...           positive
3                 10 Year Challenge - Sweet Dee Edition           positive
4  10 YEAR CHALLENGE WITH NO FILTER 47 Hilarious ...            neutral

                                         image_path
0  C:\Users\vijay\OneDrive\Desktop\meme dataset\i...
1  C:\Users\vijay\OneDrive\Desktop\meme dataset\i...
2  C:\Users\vijay\OneDrive\Desktop\meme dataset\i...
3  C:\Users\vijay\OneDrive\Desktop\meme dataset\i...
4  C:\Users\vijay\OneDrive\Desktop\meme dataset\i...
```

figure 5.2.1 Data Collection Output

figure 5.2.2 Meme Input

This dataset is part of a meme content analysis project aimed at understanding the textual and emotional characteristics of meme images. It combines image metadata, OCR-extracted text, corrected text, and sentiment classification, making it suitable for tasks like sentiment prediction, OCR benchmarking, and informal language modelling.

b) Multimodal Processing:

The multimodal processing step integrates both visual and textual features of meme data for deeper analysis. The clean text column represents pre-processed textual data refined from OCR outputs to improve accuracy. This step ensures the data is suitable for tasks like sentiment detection, classification, and NLP-based modelling.

```
                                                clean_text
0  look friend lightyear sohalikut trend play 10 ...
1  best 10 yearchallenge completed less 4 years k...
2  sam thorne strippin follow follow saw everyone...
3                 10 year challenge sweet dee edition
4  10 year challenge filter 47 hilarious 10 year ...
```

figure 5.2.3 Multimodal Meme Text Cleaning

The clean_text column contains preprocessed textual data extracted from meme images. This is likely the result of cleaning the OCR output to make it suitable for downstream tasks like sentiment analysis, classification, or NLP modeling.

c) Feature Extraction:

Feature extraction transforms textual data into numerical form, enabling machine learning models to process and analyze it effectively. Each value in the feature matrix indicates theimportance or frequency of words in a given text. This process helps identify key patterns and sentiments using techniques like TF-IDF or word embeddings.

```
Feature names: ['are' 'content' 'enjoy' 'fun' 'funny' 'love' 'memes']
Feature matrix:
 [[0.         0.         0.         0.         0.         0.79596054
  0.60534851]
 [0.62276601 0.         0.         0.62276601 0.         0.
  0.4736296 ]
 [0.         0.57735027 0.57735027 0.         0.57735027 0.
  0.        ]]
```

Figure 5.2.4: Feature Matrix Representation for Text Samples

The feature matrix converts text into numerical form for machine learning. Each row represents a text sample; each column is a word feature. Values show how important or frequent a word is in that sample. This helps models understand and classify text based on word usage. It’s commonly built using techniques like TF-IDF for NLP tasks.

d) Model Training & Evaluation:

This step involves preparing the dataset for model training by organizing text data with measurable attributes like length. It enables the model to understand relationships between text features and sentiment patterns. Such structured representation supports efficient evaluation and enhances the accuracy of NLP-based sentiment analysis models.

	text	length
0	I love memes	12
1	Memes are fun	13
2	I enjoy funny content	21

['text', 'length']

Figure 5.2.5: Text Training Features for Sentiment Prediction

This image shows a simple table with two columns : text and length .Each row contains a short sentence and it character count. The length column helps quantify text ual data for analysis.This structure is typical in NLP preprocessing using tools like pandas.

e) Sentiment Prediction:

This step demonstrates how machine learning models predict the emotional tone of text data using trained classifiers. The input text is vectorized and analyzed to determine whether its sentiment is positive, negative, or neutral. Such predictions help in understanding public opinions, meme emotions, and overall social media sentiment trends.

```python
test_text = "This meme is hilarious!"
test_vector = vectorizer.transform([test_text])
print("Predicted sentiment:", clf.predict(test_vector)[0])
```

Python

Figure 5:2:6: Sentiment Prediction Using Python Code Snippet

It supports tasks like text filtering, feature extraction, or sentiment modelling. This image shows a Python code snippet performing sentiment analysis on a sample text. The text "this meme is hilarious!" is transformed into a vector using a pre-trained vectorizer. This vector is then passed to a classifier (clf) to predict its sentiment. The output "Predicted sentiment: positive" indicates the model found the text emotionally upbeat.

5.3 DISCUSSION

The Meme Sent Review Engine project demonstrates the practical potential of multimodal artificial intelligence in analyzing complex human emotions conveyed through memes. Traditional sentiment analysis systems primarily rely on textual information and often fail to interpret sarcasm, humor, or irony embedded in meme content. By combining Natural Language Processing (NLP) and Computer Vision techniques, this system effectively overcomes such limitations, achieving a more human-like understanding of digital expressions.

During experimentation, text-based models alone achieved moderate performance, as meme captions are often short or ambiguous. Similarly, image-based models provided emotional cues but lacked contextual linguistic understanding. However, when both modalities were fused, the system achieved significant improvements in accuracy, interpretability, and emotional sensitivity, indicating that multimodal learning enables a richer representation of sentiment.

The Random Forest ensemble model exhibited robustness and stability across diverse datasets, effectively managing feature variations and minimizing the impact of noise. Performance metrics such as accuracy, precision, recall, and F1-score validated the system's capability to maintain a balanced classification. The results also indicated that multimodal fusion not only enhanced classification accuracy but also improved the system's generalization across unseen meme data.

From an application perspective, the project holds substantial commercial and research value. For businesses and marketers, the ability to automatically detect audience sentiment toward brands, products, or campaigns enables data-driven decision-making and consumer trend analysis. The insights generated from meme sentiment analysis can assist in campaign evaluation, product positioning, and social listening, allowing organizations to tailor strategies based on real-time public perception.

Furthermore, the study emphasizes the growing importance of analyzing visual-textual synergy in modern communication, paving the way for future research into emotional AI systems. Integrating deep learning architectures such as CNNs or transformers (e.g., BERT, CLIP) could further improve contextual reasoning, sarcasm detection, and multilingual analysis. This research therefore not only proves the feasibility of multimodal sentiment analysis but also establishes a framework for advancing emotion-aware computing in digital marketing and social media analytics.

Summary Of Findings

1. Text-Based Analysis: Successfully captured key emotional words but struggled with sarcasm and humour.
2. Image-Based Analysis: Extracted mood-related features (colour, expression, tone) but lacked semantic understanding.
3. Multimodal Fusion: Combined both strengths to achieve the highest accuracy and robustness.
4. Business Insight: Enabled identification of consumer sentiment trends and emotional triggers in social media memes.
5. Practical Utility: Demonstrated potential for marketing optimization, sales forecasting, and reputation management.
6. Performance Evaluation: The Random Forest classifier achieved superior accuracy compared to individual unimodal models, confirming the effectiveness of multimodal integration in handling diverse meme data.
7. Scalability and Adaptability: The system design allows easy integration with real-time data sources such as Twitter, Instagram, or Reddit, enabling continuous monitoring of consumer sentiment and trends.
8. Interpretability and Transparency: The model outputs interpretable sentiment scores and feature importance, allowing users and organizations to understand the reasoning behind predictions
9. Technological Advancement: Showcased the synergy between Natural Language Processing (NLP) and Computer Vision (CV), emphasizing the relevance of AI-driven multimodal approaches in next-generation analytics.
10. Future Scope: Encourages further development through deep learning models (CNN, BERT, CLIP) and real-time deployment, aiming to enhance precision, multilingual support, and emotion intensity detection for broader applications.

5.4 CONCLUSION OF RESULT DISCUSSION

The results confirm that meme sentiment analysis can serve as a powerful analytical tool for interpreting public opinion in the digital age. By effectively bridging the gap between textual and visual modalities, the proposed Meme Sent Review Engine demonstrates the capability to extract complex emotional cues from memes, including humor, irony, and sarcasm, which are often overlooked in conventional sentiment analysis models. This integration of multimodal processing enhances the emotional intelligence of computational systems, enabling them to better understand and interpret human expressions conveyed through digital media.

The findings further indicate that combining image-based and text-based features significantly improves classification performance and overall sentiment prediction accuracy. This hybrid approach allows deeper insights into user-generated content, enabling businesses and researchers to analyze audience attitudes, preferences, and behaviors with greater reliability. Moreover, the observed correlation between positive sentiment trends and product sales growth reinforces the model's potential application in brand monitoring, marketing optimization, and consumer engagement analytics.

In addition, the model's adaptability and scalability make it suitable for deployment across multiple platforms, offering real-time monitoring of audience sentiment from social media streams. Such capabilities can be particularly valuable for enterprises seeking to track the impact of digital marketing campaigns, measure consumer satisfaction, and anticipate market trends through automated feedback systems.

Overall, the Meme Sent Review Engine establishes a strong foundation for the advancement of AI-driven social media analysis tools. It demonstrates that machine learning models can not only interpret sentiments but also transform them into actionable business intelligence. Future research can extend this work by integrating deep learning architectures such as CNN, LSTM, and Transformer-based models (like BERT or CLIP) to capture richer contextual and visual relationships. Furthermore, coupling the system with real-time data pipelines and visualization dashboards would enable organizations to dynamically track sentiment evolution and make informed marketing decisions with higher accuracy and precision.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The Meme Sent Review Engine project has successfully demonstrated the immense potential of multimodal sentiment analysis as an advanced tool for understanding consumer emotions and brand perception through the interpretation of internet memes. By integrating both textual and visual modalities, the system transcends conventional sentiment analysis approaches that rely exclusively on textual data. It captures deeper emotional dimensions such as humor, irony, sarcasm, and cultural context, which are often communicated visually in memes. This innovation not only enriches sentiment interpretation but also strengthens emotional intelligence within digital marketing and online communication systems.

The project highlights how machine learning models can effectively process complex visual-textual cues to classify memes as positive, negative, or neutral, thus offering actionable insights into public opinion, market trends, and product perception. The framework enables organizations to make data-driven decisions, optimize advertising strategies, and enhance customer engagement through emotion-aware analytics. In essence, the system bridges the gap between artificial intelligence and human emotional understanding in the context of social media dynamics.

Future research will focus on refining model accuracy and robustness by incorporating advanced transformer-based deep learning architectures such as BERT, Vision Transformers (ViT), and CLIP. These models will enable the system to manage intricate meme structures, variations in lighting, tone, and composition, ensuring better generalization across diverse datasets. Additionally, future enhancements may involve multimodal fusion techniques that integrate facial expression recognition, emotion intensity detection, and cultural context interpretation to further improve the system's sentiment classification capabilities.

The integration of Meme Sent with popular social media platforms like Instagram, Twitter, Reddit, and YouTube will enable real-time tracking of brand-related memes and public sentiment. This real-time capability will empower analysts and marketing professionals to continuously monitor emotional trends, identify emerging issues, and respond rapidly to customer feedback. Such responsiveness will lead to improved customer relations, proactive brand management, and stronger online reputation control.

To achieve context-aware performance, the system will continue to evolve by analyzing memes based on their linguistic, situational, and cultural background. This adaptability will enhance its ability to distinguish between sarcasm, irony, humor, and genuine emotions, ensuring more nuanced and accurate sentiment predictions in dynamic online environments. Furthermore, the flexibility of the Meme Sent framework extends beyond commercial applications—it holds significant promise in political sentiment analysis, social movement tracking, entertainment feedback assessment, and public awareness campaigns.

Cloud-based deployment and GPU-accelerated processing will ensure scalability and faster computation, making the system ideal for real-time analytics and large-scale data visualization. Interactive dashboards and heatmaps will allow marketing analysts to explore sentiment patterns, keyword associations, and emotional triggers in a visually intuitive manner. This will aid in identifying consumer behavior trends and tailoring marketing strategies to align with emotional appeal.

By integrating meme sentiment data with sales analytics, market indicators, and product performance metrics, the system can assist businesses in making informed strategic decisions, improving customer experiences, and innovating product development. The resulting synergy between emotional data and business intelligence will foster deeper connections between brands and audiences.

In conclusion, the Meme Sent Review Engine not only contributes to the advancement of artificial intelligence and multimodal sentiment analysis but also establishes a comprehensive framework for understanding complex emotional dynamics in digital media. This system demonstrates how textual and visual modalities can be effectively fused to interpret user sentiments with higher precision, capturing nuances such as humor, irony, and sarcasm that are often missed in traditional models.

Beyond academic significance, the model has strong practical implications for marketing, brand monitoring, and customer engagement analysis. By evaluating audience reactions through memes and social content, businesses can make informed, emotion-driven decisions that enhance product performance and brand reputation. Furthermore, the system's adaptability makes it suitable for real-time applications, including trend forecasting, campaign optimization, and consumer behavior prediction.

As digital interactions continue to grow, the Meme Sent Review Engine provides a scalable and data-driven approach to decoding human emotions across online platforms. This research lays a foundation for future innovations in emotion-aware computing, cognitive AI, and interactive social intelligence systems, promoting deeper human–machine understanding and more empathetic digital communication in the modern world.

6.2 FUTURE SCOPE

6.2.1 Deep Learning-Based Fusion Models

The future development of the project will focus on adopting deep multimodal fusion architectures, integrating Convolutional Neural Networks (CNNs) for image understanding and Transformer models for text analysis. This will allow the system to learn complex relationships between visual and textual cues in memes.

6.2.2 Multilingual and Cross-Cultural Analysis

The system will be extended to support multiple languages and cultural meme variations. This enhancement will enable global brands to Analyse sentiment across different regions, dialects, and humour styles ,expanding the system's real-world applicability.

6.2.3 Real-Time Meme Monitoring System

A real-time meme sentiment monitoring tool will be developed to automatically collect and Analize trending memes from social media. This will allow brands to identify viral content, track reputation, and respond promptly to customer opinions.

6.2.4 Integration with Marketing Analytics Tools

The Meme Sent system can be integrated with existing marketing and business analytics platforms such as Google Analytics, Power BI, or Tableau. This integration will enable seamless visualization and interpretation of sentimental data alongside business performance metrics.

6.2.5 Emotion-Based Recommendation Systems

Future versions can include emotion-aware recommendation systems, which suggest marketing strategies, advertisements, or product improvements based on dominant meme sentiments. This can lead to more personalized and emotionally resonant campaigns.

6.2.6 Cloud and API Deployment

The project will evolve into a cloud-based service with an API interface, enabling developers and businesses to integrate meme sentiment analysis directly into their applications or analytics pipelines.

6.2.7 Dataset Expansion and Benchmarking

Larger and more diverse meme datasets will be curated for better model generalization.

Benchmarking against standard multimodal datasets will also ensure the system's credibility and research value.

6.2.8 Educational and Research Applications

The system can be utilized in academic research and education, helping students and researchers explore multimodal learning, affective computing, and artificial intelligence applications in communication studies.

6.2.9 Integration with Augmented Reality (AR) and Virtual Reality (VR)

Future work could extend the Meme Sent engine into AR/VR environments, where real-time meme sentiment could influence interactive digital experiences or virtual product marketing campaigns.

6.2.10 Global Reach and Commercialization

Ultimately, the Meme Sent Review Engine aims to become a commercially deployable AI tool capable of serving global organizations in understanding and responding to consumer emotions. Its scalability, accuracy, and adaptability make it suitable for widespread adoption in digital marketing, product analytics, and brand management.

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APPENDIX-A

Source Code:

```
<h1 style="color:white;">STEP1; Import Label Data(CSV)</h1>
import pandas as pd
label_path = r"C:\Users\vijay\OneDrive\Desktop\meme dataset\labels.csv"
df_labels = pd.read_csv(label_path)
df_labels.columns = [col.strip() for col in df_labels.columns]
 # Clean column names
print(df_labels.head())
<h2 style="color:white;">2: Import Image Data (from folder) </h2>
import os
image_folder = r"C:\Users\vijay\OneDrive\Desktop\Machine_learning\images"
image_files = os.listdir(image_folder)
df_images = pd.DataFrame({'image_name': image_files})
print(df_images.head())
<h2 style="color:white;">3: Merge Labels and Images</h2>
df = pd.merge(df_labels, df_images, on='image_name', how='inner')
df['image_path'] = df['image_name']. apply (lambda x: os.path.join(image_folder, x))
print(df.head())
<h2 style="color:white;">Tip: If you're using OCR or image embeddings later</h2>
df['image_path'] = df['image_name']. apply (lambda x: os.path.join(image_folder, x))
<h1 style="color:white;">Step 2: Clean and Preprocess Text</h1>
text_column = 'text_corrected'
<h2 style="color:white;">2. Define a cleaning function</h2>
import re
import nltk
from nltk.corpus import stopwords
from textblob import TextBlob
nltk.download('stopwords')
```

```
stop_words = set(stopwords.words('english'))
def clean_text(text):
    if not isinstance(text, str):
        return ""
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"[^a-zA-Z0-9\s]", "", text)
    return text.lower()
<h2 style="color:white;">3. Apply cleaning to your DataFrame</h2>
df['clean_text'] = df[text_column]. apply(clean_text)
print(df[['clean_text']]. head ())
<h2 style="color:white;">4. Optional: Remove stopwords</h2>
import nltk
from nltk.corpus import stopwords
# Download stopwords
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))
# Define stopword removal function
def remove_stopwords(text):
    return " “. join ([word for word in text.split() if word not in stop_words])
df['clean_text'] = df['text_corrected']. apply(clean_text). apply(remove_stopwords)
print(df[['clean_text']]. head ())
print(df.columns.tolist())
print(df.columns.tolist())
print(df['overall_sentiment'].value_counts())
df_balanced = df.groupby('overall_sentiment'). apply(
    lambda x: x.sample(n=min(len(x), 500), random_state=42)
).reset_index(drop=True)
<h3 style="color:white;">Split the Data</h3>
from sklearn.model_selection import train_test_split
```

```
# Vectorize the balanced dataset
from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(df_balanced['clean_text'])
y = df_balanced['overall_sentiment']
# Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
<h3 style="color:white;">Train and Predict</h3>
from sklearn.linear_model import LogisticRegression
# Train the model
clf = LogisticRegression(max_iter=1000)
clf.fit(X_train, y_train)
# Predict on test set
y_pred = clf.predict(X_test)
<h3 style="color:white;">Graphical Comparison</h3>
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Create comparison DataFrame
comparison_df = pd.DataFrame({
    'Actual': y_test.values,
    'Predicted': y_pred
})
# Confusion matrix heatmap
label_counts = comparison_df.groupby(['Actual', 'Predicted']).size().unstack(fill_value=0)
plt.figure(figsize=(6, 5))
sns.heatmap(label_counts, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix Heatmap")
plt.ylabel("Actual Sentiment")
```

```
plt.xlabel("Predicted Sentiment")plt.show()
# Bar chart of Actual vs Predicted
plt.figure(figsize=(8, 5))
sns.countplot(data=comparison_df, x='Actual', hue='Predicted', palette='Set2')
plt.title("Actual vs Predicted Sentiment")
plt.ylabel("Count")
plt.xlabel("Actual Sentiment")
plt.legend(title="Predicted")
plt.show()
<h1 style="color:white;">Step 3: Convert Text to Features (TF-IDF)</h1>
<h2 style="color:white;">1.Bag of Words</h2>
print(df.columns)
<h2 style="color:white;">TF-IDF Vectorization</h2>
from sklearn.feature_extraction.text import TfidfVectorizer
# Sample texts
texts = ["I love memes", "Memes are fun", "I enjoy funny content"]
# Initialize and fit TF-IDF vectorizer
vectorizer = TfidfVectorizer()
X_tfidf = vectorizer.fit_transform(texts)
# Output feature names and matrix
print("Feature names:", vectorizer.get_feature_names_out())
print("Feature matrix:\n", X_tfidf.toarray())
<h2 style="color:white;">2.Word Embeddings (Using spaCy) </h2>
import spacy
# Load the spaCy model (make sure it's installed)
nlp = spacy.load("en_core_web_md")
# Sample text
text = "Memes are fun and expressive"
doc = nlp(text)
```

```
# Convert to feature vector by averaging token vectors
feature_vector = doc.vector
# Output the shape of the embedding
print("Embedding vector shape:", feature_vector.shape)
<h2 style="color:white;">4.Custom Feature Extraction (e.g., Text Length, Sentiment)</h2>
import pandas as pd
from textblob import TextBlob
# Sample texts
texts = ["I love memes", "Memes are fun", "I enjoy funny content"]
# Create DataFrame
df = pd.DataFrame(texts, columns=["text"])
# Add text length feature
df["length"] = df["text"].apply(len)
# Preview DataFrame
print(df)
# Clean column names
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')
# Preview column names
print(df.columns.tolist())
print(df.columns)
df.columns = df.columns.str.strip().str.lower()
from textblob import TextBlob
df['polarity'] = df['text'].apply(lambda x: TextBlob(x).sentiment.polarity)
<h2 style="color:white;">Custom Features (Length, Sentiment)</h2>
def get_sentiment(p):
    if p > 0.01:
        return 'positive'
    elif p < -0.01:
        return 'negative'
```

```
    else: return 'neutral'
# Apply the function to the 'polarity' column
df['sentiment'] = df['polarity'].apply(get_sentiment)
# Display sentiment distribution
print(df['sentiment'].value_counts())
print(df.columns.tolist())
df['clean_text'] = df['text'].astype(str).str.lower().str.replace(r'[^a-z\s]', '', regex=True)
<h2 style="color:white;">Step 4:Sentiment Analysis with VADER</h2>
from nltk.sentiment import SentimentIntensityAnalyzer
nltk.download('vader_lexicon')
sia = SentimentIntensityAnalyzer()
def vader_sentiment(text):
    score = sia.polarity_scores(text)['compound']
    if score > 0.05:
        return 'positive'
    elif score < -0.05:
        return 'negative'
    else:
        return 'neutral'
df['sentiment'] = df['clean_text'].apply(vader_sentiment)
print(df['sentiment'].value_counts())
<h2 style="color:white;">Step 5: Balance Dataset for Binary Classification</h2>
# Step 1: Check how many sentiment classes exist
unique_classes = df['sentiment'].unique()
print("Unique sentiment classes:", unique_classes)
# Step 2: Filter for binary classification (positive vs negative)
df_filtered = df[df['sentiment'].isin(['positive', 'negative'])]
# Step 3: Check class counts
class_counts = df_filtered['sentiment'].value_counts()
```

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print("Class counts:\n", class_counts)
# Step 4: Balance the dataset if both classes are present
if len(class_counts) >= 2:
    min_count = class_counts.min()
    df_balanced = (
        df_filtered
        .groupby('sentiment')
        .apply(lambda x: x.sample(min_count, random_state=42))
        .reset_index(drop=True)
    )
    print(" Balanced class counts:\n", df_balanced['sentiment'].value_counts())
else:
    print("Not enough class diversity to train a model.")
print(df['sentiment'].value_counts(dropna=False))
print(df['clean_text'].head())
new_texts = [
    "I love memes",              # positive
    "Memes are fun",             # positive
    "I enjoy funny content",     # positive
    "This meme is boring",       # negative
    "I hate this meme",          # negative
    "Meh, it's okay I guess",    # neutral
    "Not really funny",          # negative
    "Absolutely hilarious!",     # positive
    "I don't get it",            # neutral
]
df = pd.DataFrame({'clean_text': new_texts})
df['sentiment'] = df['clean_text'].apply(vader_sentiment)
print(df['sentiment'].value_counts())
```

```
# Filter for binary classification
df_filtered = df[df['sentiment'].isin(['positive', 'negative'])]
# Check class counts
class_counts = df_filtered['sentiment'].value_counts()
# Balance the dataset
if len(class_counts) >= 2:
    min_count = class_counts.min()
    df_balanced = (
        df_filtered
        .groupby('sentiment')
        .apply(lambda x: x.sample(min_count, random_state=42))
        .reset_index(drop=True)
    )
    print("Balanced class counts:\n", df_balanced['sentiment'].value_counts())
else:
    print(" Not enough class diversity to train a model.")
# Vectorize and train
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(df_balanced['clean_text'])
y = df_balanced['sentiment']
<h2 style="color:white;">Step 6: Train Sentiment Classifier</h2>
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# Train the model
clf = LogisticRegression(max_iter=1000)
clf.fit(X_train, y_train)
```

```
# Predict and evaluate
y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred))
<h2 style="color:white;">Step 7: Model Evaluation & Visualization</h2>
from sklearn.metrics import ConfusionMatrixDisplay
ConfusionMatrixDisplay.from_estimator(clf, X_test, y_test)
<h2 style="color:white;">Visualize TF-IDF feature importance</h2>
import numpy as np
feature_names = vectorizer.get_feature_names_out()
coefs = clf.coef_[0]
top_features = np.argsort(coefs)[-10:]
print("Top positive indicators:", feature_names[top_features])
 <h3 style="color:white;">Deploy the Model</h3>
import joblib
joblib.dump(clf, 'sentiment_model.pkl')
joblib.dump(vectorizer, 'tfidf_vectorizer.pkl')
import joblib
# Load the trained model and vectorizer
clf = joblib.load('sentiment_model.pkl')
vectorizer = joblib.load('tfidf_vectorizer.pkl')
# New input text
test_text = "This meme is hilarious!"
# Transform using the same vectorizer
test_vector = vectorizer.transform([test_text])
# Predict sentiment
predicted_label = clf.predict(test_vector)[0]
print("Predicted sentiment:", predicted_label)
import matplotlib.pyplot as plt
# Get prediction probabilities
```

```
probabilities = clf.predict_proba(test_vector)[0]

class_labels = clf.classes_

# Plot

plt.figure(figsize=(6, 4))

bars = plt.bar(class_labels, probabilities, color=['red' if label == 'negative' else 'green' for
label in class_labels])

plt.title(f"Sentiment Prediction for: \"{test_text}\"")

plt.ylabel("Probability")

plt.ylim(0, 1)

# Annotate bars

for bar, prob in zip(bars, probabilities):

    plt.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.02,

             f"{prob:.2f}", ha='center', va='bottom', fontsize=10)

plt.tight_layout()

plt.show()

<h3 style="color:white;">To Test with New Input</h3>

test_text = "This meme is hilarious!"

test_vector = vectorizer.transform([test_text])

print("Predicted sentiment:", clf.predict(test_vector)[0])
```